Structure Selection Theorem

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I. TOWARDS GLOBAL OPTIMALITY: CAN ANY 3D STAR BEAT BCC IN Λ_S?

Problem Formulation

Let $S \subset \mathbb{S}^2_{q_0} = \{k \in \mathbb{R}^3 : \|k\| = q_0\}$ be a finite star set satisfying inversion symmetry ($k \in S \Rightarrow -k \in S$). Define the weighted quartic resonance invariant

$$\Lambda(S) := \sum_{\substack{(k_1,k_2,k_3,k_4)\in S^4 \\ k_1+k_2+k_3+k_4=0}} F(\alpha_{1234}), \qquad F(\alpha) \geq 0.$$

For $F \equiv 1$, $\Lambda(S)$ reduces (up to convention) to the additive energy on the sphere:

$$E(S) := \#\{(k_1, k_2, k_3, k_4) \in S^4 : k_1 + k_2 = k_3 + k_4\}.$$

Unconstrained maximization of $\Lambda(S)$ is ill-posed: increasing $|S|$ typically increases $\Lambda(S)$. Therefore, meaningful optimality questions require a restriction of the admissible class or a normalization.

Three Meaningful Optimality Statements

(A) Fixed-size optimality. Fix $N = |S|$. Define

$$\Lambda^{\max}_N := \max\{\Lambda(S) : S \subset \mathbb{S}^2_{q_0},\ S = -S,\ |S| = N\}.$$

A concrete target is $N = 12$:

$$\Lambda^{\max}_{12} \stackrel{?}{=} \Lambda(S_{\mathrm{BCC}}),$$

where S_{BCC} denotes the 12-vector BCC star on the first shell.

(B) Bravais first-shell optimality. Restrict S to be the set of shortest nonzero reciprocal lattice vectors of a 3D Bravais lattice. Then $|S|$ is not arbitrary, and one may ask whether BCC maximizes Λ within this class:

$$\Lambda(S_{\mathrm{BCC}}) \geq \Lambda(S) \quad \forall S \in \mathcal{S}_{\mathrm{Bravais},1}.$$

(C) Normalized optimality. Define a normalized functional, e.g.

$$\widehat{\Lambda}(S) := \frac{\Lambda(S)}{|S|^3} \quad \text{or} \quad \widetilde{\Lambda}(S) := \frac{\Lambda(S)}{|S|^2},$$

and maximize $\widehat{\Lambda}$ or $\widetilde{\Lambda}$ over all antipodal stars.

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Reduction to Additive-Energy Optimization (Unweighted Case)

For $F \equiv 1$, the objective becomes the additive energy $E(S)$. By Cauchy–Schwarz,

$$E(S) = \sum_v r_S(v)^2, \qquad r_S(v) := \#\{(k_1, k_2) \in S^2 : k_1 + k_2 = v\},$$

and

$$E(S) \geq \frac{\left(\sum_v r_S(v)\right)^2}{|\{v : r_S(v) > 0\}|} = \frac{|S|^4}{|\mathrm{Sum}(S)|}, \quad \mathrm{Sum}(S) := \{k_1 + k_2 : k_i \in S\}.$$

Thus, maximizing $E(S)$ favors configurations with a small number of distinct pair-sums, i.e. strong pair-sum degeneracy.

Geometric Mechanism Behind BCC

The BCC star exhibits unusually high degeneracy of pair-sums and four-wave closures under the constraint that all vectors lie on a single radius and the set is antipodal. This produces a large contribution to $r_S(v)^2$ and hence to $E(S)$ (and $\Lambda(S)$ for $F \geq 0$).

Proof Strategy for (A) and (B)

Step 1 (Classification of admissible $|S|$). For Bravais first-shell stars, enumerate possible shell multiplicities (e.g. $|S| \in \{6, 8, 12, \dots\}$) and reduce comparison to finitely many symmetry types.

Step 2 (Upper bounds via sumset size). Derive lower bounds on $|\mathrm{Sum}(S)|$ from spherical geometry constraints, implying upper bounds on $E(S)$ and hence on $\Lambda(S)$ for $F \geq 0$.

Step 3 (Sharpness). Show that BCC saturates the bound (or is the unique maximizer) within the admissible class. For $N = 12$, this becomes a finite extremal problem on antipodal spherical codes.

Step 4 (Weighted kernels). For general $F(\alpha) \geq 0$, use

$$\Lambda(S) \leq \left(\sup_\alpha F(\alpha)\right) E(S), \qquad \Lambda(S) \geq \left(\inf_\alpha F(\alpha)\right) E(S),$$

or more refined angle-class decompositions, to transfer the extremal statement from $E(S)$ to $\Lambda(S)$.

Implication

If either (A) with $N = 12$ or (B) over Bravais first-shell stars is proven, then BCC is not only robust among cubic candidates but extremal within a substantially larger admissible class. This would upgrade the result from a hierarchy theorem to an optimality principle for 3D shell-degenerate selection.

II. SELECTION FUNCTIONAL FOR BLOCK-COPOLYMER MICROPHASES

We consider a standard Landau–Brazovskii description of a diblock-copolymer melt near the order–disorder transition (ODT), in which the dominant density mode softens on a finite wavenumber shell $|q| \approx q_0$. After integrating out fast modes, the coarse-grained free energy for the relevant scalar order parameter Ψ takes the form

$$\mathcal{F}[\Psi] = \frac{1}{2}\int_q \Gamma_2(q)\,|\Psi_q|^2 + \frac{1}{4!}\int_{q_1 \cdots q_4} (2\pi)^d \delta\Big(\sum_{i=1}^4 q_i\Big)\,\Gamma_4(q_1, q_2, q_3, q_4)\prod_{i=1}^4 \Psi_{q_i} + \frac{\gamma}{6}\int d^d x\,\Psi^6 + \cdots, \qquad \gamma > 0.$$

We restrict to the dominant shell $|q_i| = q_0$ and assume isotropy so that the quartic vertex depends only on the internal angles:

$$\Gamma_4(q_1, q_2, q_3, q_4)\Big|_{|q_i|=q_0} = F(\alpha_{1234}), \qquad F(\alpha) \geq 0,$$

where α_{1234} denotes the set of angles between the participating shell vectors.

III. STARS FOR LAM, HEX, AND BCC

We approximate the ordered phases by equal-amplitude stars on the shell:

$$\Psi(x) = \sum_{k \in S} A\, e^{ik\cdot x}, \qquad |k| = q_0, \qquad A \in \mathbb{R}.$$

The candidate stars for the classical microphases are:

Lamellae (LAM)

$$S_{\mathrm{LAM}} = \{\pm k_1\}, \qquad |S_{\mathrm{LAM}}| = 2.$$

Hexagonal cylinders (HEX)

A 2D hexagonal star in a plane (six directions separated by 60°):

$$S_{\mathrm{HEX}} = \{\pm k_1, \pm k_2, \pm k_3\}, \qquad k_1 + k_2 + k_3 = 0, \qquad |S_{\mathrm{HEX}}| = 6.$$

BCC spheres (BCC)

The 12-vector cuboctahedral star:

$$S_{\mathrm{BCC}} = \frac{q_0}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \qquad |S_{\mathrm{BCC}}| = 12.$$

IV. DEFINITION OF THE SELECTION FUNCTIONAL Λ_S

Under the equal-amplitude ansatz, the quartic contribution reduces to

$$F_{4,S} = \frac{A^4}{4}\, \Lambda_S,$$

where the structural functional is the weighted sum over resonant quartets

$$\Lambda_S := \sum_{\substack{(k_1,k_2,k_3,k_4) \in S^4 \\ k_1+k_2+k_3+k_4=0}} F(\alpha_{1234}).$$

For a local vertex $F(\alpha) \equiv \text{const}$, Λ_S reduces to a pure resonance count (additive energy on the star).

V. ANGLE-CLASS REDUCTION

For isotropic kernels, only finitely many angle-classes contribute for each star. We therefore decompose

$$\Lambda_S = \sum_{c \in \mathcal{C}(S)} M_c(S)\, F_c,$$

where

- $\mathcal{C}(S)$ is the finite set of angle-classes realized by resonant quartets on S,
- $M_c(S) \in \mathbb{N}$ are purely combinatorial multiplicities,
- F_c are the kernel values for the corresponding quartet geometry (angles).

This is the minimal reduction needed to connect a microscopic polymer vertex to phase selection.

VI. EXPLICIT MINIMAL PARAMETRIZATION

A practical and widely used reduction is to express the quartic kernel by a few representative configurations that occur in LAM/HEX/BCC resonant quartets. We introduce:

$$F_{\parallel} := F(\text{collinear pairing}), \qquad F_{60} := F(\text{a } 60^\circ \text{ pairing geometry}), \qquad F_{90} := F(\text{a } 90^\circ \text{ pairing geometry}),$$

and additional classes if needed (e.g. tetrahedral-like quartets for 3D stars). Then each phase has

$$\Lambda_{\mathrm{LAM}} = M_{\parallel}(\mathrm{LAM})\, F_{\parallel},$$

$$\Lambda_{\mathrm{HEX}} = M_{\parallel}(\mathrm{HEX})\, F_{\parallel} + M_{60}(\mathrm{HEX})\, F_{60},$$

$$\Lambda_{\mathrm{BCC}} = M_{\parallel}(\mathrm{BCC})\, F_{\parallel} + M_{60}(\mathrm{BCC})\, F_{60} + M_{90}(\mathrm{BCC})\, F_{90} + \cdots .$$

All microscopic information enters only through a small set of numbers F_c, while all structural information enters only through the integers $M_c(S)$.

VII. SELECTION CRITERION AND PHASE BOUNDARIES

Within the quartic-stabilized Brazovskii regime, structure selection is controlled by the effective quartic coefficient, equivalently by Λ_S (up to universal prefactors). We write the reduced Landau free energy for each candidate phase as

$$f_S(A) = \frac{r}{2}A^2 + \frac{\Lambda_S}{4}A^4 + \frac{\gamma}{6}A^6, \qquad \gamma > 0.$$

Assuming sextic neutrality, the selected phase is the one minimizing f_S, which is determined by the ordering of Λ_S:

$$S^{=\arg\min_{S\,2\,f\mathrm{LAM;HEX;BCC}g}\,\Lambda_S}.$$

Phase boundaries are given by equality of the corresponding quartic functionals:

$$\Lambda_{\mathrm{LAM}} = \Lambda_{\mathrm{HEX}}, \qquad \Lambda_{\mathrm{HEX}} = \Lambda_{\mathrm{BCC}}, \qquad \Lambda_{\mathrm{LAM}} = \Lambda_{\mathrm{BCC}}.$$

In the reduced parametrization these become linear conditions in the kernel values F_c, e.g.

$$\big(M_{\parallel}(\mathrm{LAM}) - M_{\parallel}(\mathrm{HEX})\big)F_{\parallel} = M_{60}(\mathrm{HEX})F_{60},$$

and similarly for the other boundaries.

VIII. PRACTICAL PROGRAM FOR A QUANTITATIVE BCP PHASE DIAGRAM

To obtain a quantitative phase diagram in terms of polymer parameters (e.g. volume fraction f and χN), one computes the isotropic quartic vertex on the shell, extracts the finite set of kernel values $\{F_c(f,\chi N)\}$, and evaluates

$$\Lambda_S(f,\chi N) = \sum_c M_c(S)\, F_c(f,\chi N).$$

The phase at given parameters is predicted by the minimal Λ_S, and the phase boundaries follow from pairwise equalities of Λ_S.

This reduction makes explicit which part of the phase diagram is universal (the integers $M_c(S)$) and which part is material-specific (the kernel values F_c).

IX. ANGLE-CLASS MULTIPLICITIES FOR LAM/HEX/BCC AND LINEAR PHASE BOUNDARIES

We work on a single instability shell $|\mathbf{k}| = q_0$ and use equal-amplitude stars

$$(x) = \sum_{k\in S} A\, e^{ik\cdot x}; \qquad A \in \mathbb{R}; \qquad S = -S:$$

Assume an isotropic quartic vertex restricted to the shell depends on the angle between two incoming shell vectors,

$$\Gamma_4(k_1,k_2,k_3,k_4)\Big|_{|k_i|=q_0} = F(\theta_{12}); \qquad \theta_{12} := \angle(k_1,k_2); \qquad F(\theta) \neq 0;$$

for resonant quartets $k_1 + k_2 + k_3 + k_4 = 0$.

Define the (ordered) resonance functional

$$\mathcal{R}_S := \sum_{\substack{(k_1,k_2,k_3,k_4)\in S^4\\ k_1+k_2+k_3+k_4=0}} F(\theta_{12}):$$

Then the quartic contribution is

$$F_{4;S} = \frac{A^4}{4}\,\mathcal{R}_S:$$

Amplitude Normalization Across Different Stars

Because different phases use different numbers of modes $N_S := |S|$, it is convenient to compare phases at fixed quadratic norm

$$\sum_{k\in S} |A_k|^2 = N_S A^2 = \rho^2:$$

Equivalently,

$$A = \frac{\rho}{\sqrt{N_S}}:$$

Then

$$F_{4;S} = \frac{\rho^4}{4}\,\bar{\mathcal{R}}_S; \qquad \bar{\mathcal{R}}_S := \frac{\mathcal{R}_S}{N_S^2}:$$

Thus selection among phases is controlled by the normalized functional $\bar{\mathcal{R}}_S$.

Stars

Lamellae (LAM):

$$S_{\mathrm{LAM}} = \{\pm k_1\}; \qquad N_{\mathrm{LAM}} = 2:$$

Hexagonal cylinders (HEX) (planar hexagonal star):

$$S_{\mathrm{HEX}} = \{\pm k_1, \pm k_2, \pm k_3\}; \qquad k_1 + k_2 + k_3 = 0; \qquad N_{\mathrm{HEX}} = 6:$$

BCC spheres (BCC) (cuboctahedral star):

$$S_{\mathrm{BCC}} = \frac{q_0}{\sqrt{2}}\{(\pm 1,\pm 1,0),(\pm 1,0,\pm 1),(0,\pm 1,\pm 1)\}; \qquad N_{\mathrm{BCC}} = 12:$$

Finite Angle Classes

For these stars, the only angles that occur between star vectors are

$$2 \{0^\circ, 60^\circ, 90^\circ, 120^\circ, 180^\circ\}.$$

De ne

$$F_0 := F(0^\circ); \quad F_{60} := F(60^\circ); \quad F_{90} := F(90^\circ); \quad F_{120} := F(120^\circ); \quad F_{180} := F(180^\circ):$$

Then

$$_S = \sum M_S(\theta) F_\theta; \qquad \bar{}_S = \sum \overline{M}_S(\theta) F_\theta; \qquad \overline{M}_S(\theta) := \frac{M_S(\theta)}{N_S^2}:$$

Explicit Multiplicities (Ordered Resonant Quartets)

The ordered resonance counts grouped by the angle $_{12}$ are:

a. LAM (N = 2).

$$_{LAM} = 2F_0 + 4F_{180}; \qquad \bar{}_{LAM} = \frac{1}{2}F_0 + 1 \quad F_{180}:$$

b. HEX (N = 6).

$$_{HEX} = 6F_0 + 24F_{60} + 24F_{120} + 36F_{180};$$

$$\bar{}_{HEX} = \frac{1}{6}F_0 + \frac{2}{3}F_{60} + \frac{2}{3}F_{120} + 1 \quad F_{180}:$$

c. BCC (N = 12).

$$_{BCC} = 12F_0 + 96F_{60} + 96F_{90} + 192F_{120} + 144F_{180};$$

$$\bar{}_{BCC} = \frac{1}{12}F_0 + \frac{2}{3}F_{60} + \frac{2}{3}F_{90} + \frac{4}{3}F_{120} + 1 \quad F_{180}:$$

Linear Phase-Boundary Conditions

Within the quartic-stabilized Landau{Brazovskii description, the selected phase among fLAM; HEX; BCCg is predicted by

$$S = \arg\min_S \bar{}_S;$$

and phase boundaries are determined by equalities of $\bar{}_S$.

d. LAM{HEX boundary.

$$\bar{}_{LAM} = \bar{}_{HEX} \quad \Leftrightarrow \quad \frac{1}{2} \quad \frac{1}{6} F_0 = \frac{2}{3}F_{60} + \frac{2}{3}F_{120}:$$

Equivalently,

$$\frac{1}{3}F_0 = \frac{2}{3}(F_{60} + F_{120}) \quad \Leftrightarrow \quad F \quad _0 = 2(F_{60} + F_{120}):$$

e. HEX{BCC boundary.

$$\bar{}_{HEX} = \bar{}_{BCC} \quad \Leftrightarrow \quad \frac{1}{6} \quad \frac{1}{12} F_0 + \frac{2}{3}F_{120} = \frac{2}{3}F_{90} + \frac{4}{3}F_{120}:$$

Equivalently,

$$\frac{1}{12}F_0 = \frac{2}{3}F_{90} + \frac{2}{3}F_{120} \quad \Leftrightarrow \quad F \quad _0 = 8(F_{90} + F_{120}):$$

f. LAM{BCC boundary.

$$\bar{\chi}_{\mathrm{LAM}} = \bar{\chi}_{\mathrm{BCC}} \quad \Leftrightarrow \quad \frac{1}{2} \quad \frac{1}{12} F_0 = \frac{2}{3}F_{60} + \frac{2}{3}F_{90} + \frac{4}{3}F_{120}.$$

Equivalently,

$$\frac{5}{12}F_0 = \frac{2}{3}(F_{60} + F_{90} + 2F_{120}) \quad \Leftrightarrow \quad 5F_{0} = 8(F_{60} + F_{90} + 2F_{120}).$$

Interpretation

All material dependence is contained in the nite set of kernel values $\{F_0; F_{60}; F_{90}; F_{120}; F_{180}\}$, which may be computed from the microscopic block-copolymer vertex. All structural dependence is contained in the universal rational coe cients in $\bar{\chi}_S$ above.

In particular, the LAM/HEX/BCC selection reduces to comparing three linear forms in $\{F\}$.

X. MICROSCOPIC ORIGIN OF THE ANGLE-DEPENDENT QUARTIC KERNEL IN DIBLOCK COPOLYMERS

We outline a standard Edwards{Leibler route that yields the Landau{Brazovskii functional for diblock melts. Consider an incompressible AB diblock melt of Gaussian chains (segment length b, polymerization index N), with volume fraction f of A segments and (1 f) of B segments. Let $\phi_A(x)$; $\phi_B(x)$ be local volume fractions with incompressibility $\phi_A + \phi_B = 1$. Introduce the composition order parameter

$$\psi(x) := \phi_A(x) \quad \phi_B(x).$$

After a Hubbard{Stratonovich transformation and integrating out polymer degrees of freedom, one obtains an e ective action for ψ whose vertices are connected cumulants of the single-chain composition density.

Single-Chain Composition Density and Cumulants

Parameterize a Gaussian chain by contour $s \in [0,1]$ and position $\mathbf{r}(s)$. De ne the single-chain microscopic composition density in Fourier space:

$$\hat{\psi}_{\mathbf{q}} := \int_0^1 ds\, \sigma(s)\, e^{i\mathbf{q}\cdot\mathbf{r}(s)}; \qquad \sigma(s) = \begin{cases} +1, & s \in [0,f], \\ -1, & s \in (f,1]. \end{cases}$$

The n-point vertex functions on the disordered reference state are proportional to the connected cumulants

$$\Gamma_n(\mathbf{q}_1,\dots,\mathbf{q}_n) \propto \left\langle \hat{\psi}_{\mathbf{q}_1} \cdots \hat{\psi}_{\mathbf{q}_n} \right\rangle_c \quad \text{with} \quad \sum_{i=1}^{n} \mathbf{q}_i = 0,$$

where $\langle\cdots\rangle_c$ denotes the connected part computed in the ideal-chain measure.

XI. EXPLICIT GAUSSIAN-CHAIN FORMULA FOR THE QUARTIC VERTEX [4]

For Gaussian chains, multi-point averages reduce to contour integrals with an exponential kernel. A convenient exact representation is obtained by writing (before taking the connected part)

$$\left\langle \prod_{a=1}^{4} e^{i\mathbf{q}_a\cdot\mathbf{r}(s_a)} \right\rangle = \exp\left(\frac{b^2 N}{6} \sum_{a<b} \mathbf{q}_a \cdot \mathbf{q}_b\, |s_a - s_b| \right).$$

Hence the (unreduced) four-point moment is

$$M_4(q_1;q_2;q_3;q_4;f) := \int_0^1 ds_1 \cdots \int_0^1 ds_4 \prod_{a=1}^{4} (s_a) \quad \exp\left[\frac{b^2 N}{6} \sum_{a<b} q_a \quad q_b j s_a \quad s_b j \right] :$$

The connected cumulant is obtained by subtracting Wick-type products of lower moments:

$$C_4(q_1;q_2;q_3;q_4;f) = M_{\ 4}(q_1;q_2;q_3;q_4;f) \sum_{\text{pairings}} M_2(q_i;q_j;f) M_{\ 2}(q_k;q_l;f);$$

where the sum runs over the three distinct pairings of $f1;2;3;4g$ and

$$M_2(q;\ q;f) = \int_0^1 ds \int_0^1 ds^0 \ (s) \ (s \quad {}^0) e^{\ xjs \ s \quad {}^0 j}; \qquad x := \frac{b^2 N}{6} q^2 :$$

In the Landau expansion, $_4$ is proportional to C_4 (up to a conventional prefactor depending on density normalization).

XII. SHELL RESTRICTION AND ANGLE-DEPENDENT KERNEL F( )

Near the ODT, the quadratic vertex $_2(q)$ develops a minimum at $jqj = q_0$ (nite-q instability). Restricting all external legs to the shell,

$$jq_1j = jq_{\ 2}j = jq_{\ 3}j = jq_{\ 4}j = q_{\ 0}; \qquad q_1 + q_2 + q_3 + q_4 = 0;$$

isotropy implies that $_4$ depends on a nite set of scalar invariants. For the resonance quartets relevant to the star-amplitude theory (pair-sum closure), it is su cient to evaluate the special quartet geometry

$$(q_1;q_2;q_3;q_4) = (k;\ k;p;\ p); \qquad jkj = jpj = q \quad {}_0; \qquad k \quad p = q_0^2 \cos \ :$$

We then de ne the angle-dependent quartic kernel by

$$F(\ ;f) := \quad {}_4(k;\ k;p;\ p) \Big|_{jkj=jpj=q \quad {}_0} :$$

Equivalently (up to prefactors),

$$F(\ ;f) \ / \ C \quad {}_4(k;\ k;p;\ p;f); \qquad \cos \ = \ \frac{k \quad p}{q_0^2} :$$

Thus $F(\ ;f)$ is fully determined by the explicit contour integrals above.

XIII. PRACTICAL EVALUATION AT DISCRETE ANGLES FOR LAM/HEX/BCC

For the LAM/HEX/BCC stars, only the discrete angles

$$2 \ f0 \quad ;60 \ ;90 \ ;120 \ ;180 \ g$$

occur between star vectors. De ne

$$F_0(f) := F(0 \quad ;f); \quad F_{\ 60}(f) := F(60 \quad ;f); \quad F_{\ 90}(f) := F(90 \quad ;f); \quad F_{\ 120}(f) := F(120 \quad ;f); \quad F_{\ 180}(f) := F(180 \quad ;f):$$

Each of these is obtained by inserting the corresponding cos into the quartet $(k;\ k;p;\ p)$ and evaluating $C_{\ 4}$.

Algorithmic Form (Contour Ordering Reduction)

The integral for M_4 can be reduced to a finite sum over contour orderings. Let σ be a permutation of $\{1,2,3,4\}$ and restrict to the simplex

$$0 \le s_{\sigma(1)} \le s_{\sigma(2)} \le s_{\sigma(3)} \le s_{\sigma(4)} \le 1.$$

On each simplex, $|s_a - s_b|$ becomes linear, and

$$M_4 = \sum_{\sigma} \int_{2S_4} \int_{0 \le s_{\sigma(1)} \le \cdots \le s_{\sigma(4)} \le 1} ds_1 \cdots ds_4 \prod_{a=1}^{4} \rho(s_a) \exp\left(\frac{b^2 N}{6} \sum_{a<b} q_a q_b (s_{\sigma(b)} - s_{\sigma(a)}) \right).$$

Since $\rho(s)$ is piecewise constant with a single breakpoint at $s = f$, each simplex integral further splits into finitely many regions depending on whether each s_a lies in $[0,f]$ or $(f,1]$. Therefore $F(\rho, f)$ is computable exactly as a finite sum of elementary exponentials in $\chi_0 := \frac{b^2 N}{6} q_0^2$.

XIV. CLOSING THE LAM/HEX/BCC PHASE BOUNDARIES IN TERMS OF $F_\rho(f)$

Using the normalized quartic functionals derived previously,

$$\bar{\gamma}_{\mathrm{LAM}} = \frac{1}{2} F_0(f) + F_{180}(f),$$

$$\bar{\gamma}_{\mathrm{HEX}} = \frac{1}{6} F_0(f) + \frac{2}{3} F_{60}(f) + \frac{2}{3} F_{120}(f) + F_{180}(f),$$

$$\bar{\gamma}_{\mathrm{BCC}} = \frac{1}{12} F_0(f) + \frac{2}{3} F_{60}(f) + \frac{2}{3} F_{90}(f) + \frac{4}{3} F_{120}(f) + F_{180}(f),$$

the mean-field selection among LAM/HEX/BCC reduces to

$$S^*(f) = \arg\min_{S \in \{\mathrm{LAM},\mathrm{HEX},\mathrm{BCC}\}} \bar{\gamma}_S(f),$$

and the phase boundaries are the linear equalities:

LAM{HEX

$$\bar{\gamma}_{\mathrm{LAM}} = \bar{\gamma}_{\mathrm{HEX}} \iff F_0(f) = 2\,F_{60}(f) + F_{120}(f).$$

HEX{BCC

$$\bar{\gamma}_{\mathrm{HEX}} = \bar{\gamma}_{\mathrm{BCC}} \iff F_0(f) = 8\,F_{90}(f) + F_{120}(f).$$

LAM{BCC

$$\bar{\gamma}_{\mathrm{LAM}} = \bar{\gamma}_{\mathrm{BCC}} \iff 5F_0(f) = 8\,F_{60}(f) + F_{90}(f) + 2F_{120}(f).$$

XV. WHERE N ENTERS AND WHY THE ANGLE KERNEL IS MOSTLY A FUNCTION OF f

At the level of the Landau expansion about the disordered melt, the angle dependence of the quartic vertex on the instability shell is controlled primarily by the single-chain statistics, hence by f and the dimensionless shell variable $x_0 = \frac{b^2 N}{6} q_0^2$. The location q_0 is determined by the quadratic vertex $_2(q)$ and thus depends on N through the proximity to the ODT. Operationally, one computes $q_0(f; N)$ from $\,^0_2(q_0) = 0$ and evaluates F (f) at $x_0(f; N)$.

This closes a quantitative route from microscopic parameters (f; N) to the LAM/HEX/BCC phase boundaries above.

XVI. QUADRATIC VERTEX IN RPA AND THE DETERMINATION OF $q_0(f; N)$

We summarize the standard random-phase approximation (RPA) for an incompressible AB diblock melt and extract the nite-q instability scale q_0.

A. Dimensionless Variables

Let $R_g^2 = b^2 N/6$ and de ne

$$x := q^2 R_g^2 :$$

We use the contour parameter $s \in [0; 1]$ with A-block on $[0; f]$ and B-block on $(f; 1]$.

B. Single-Chain Block Correlators

For an ideal Gaussian chain,

$$e^{iq\,(r(s)\, r(s'))} = \exp(-x|s - s'|) :$$

De ne the block-block correlators on the shell variable x:

$$_{AA}(x; f) := \int_0^f ds \int_0^f ds' e^{-x|s-s'|} = \frac{2}{x^2}\,[x - 1 + e^{-fx}] ;$$

$$_{BB}(x; f) := \int_f^1 ds \int_f^1 ds' e^{-x|s-s'|} = \frac{2}{x^2}\,[(1-f)x - 1 + e^{-(1-f)x}] ;$$

$$_{AB}(x; f) := \int_0^f ds \int_f^1 ds' e^{-x(s'-s)} = \frac{(1 - e^{-fx})(1 - e^{-(1-f)x})}{x^2} :$$

For the composition mode $= _A - _B$, the single-chain composition correlator is

$$g(x; f) := {}_{AA}(x; f) + {}_{BB}(x; f) - 2\,{}_{AB}(x; f):$$

C. RPA Quadratic Vertex

Up to an overall (convention-dependent) prefactor, the quadratic vertex for the composition mode takes the form

$$_2(q) \propto {}_2(x; f; N) = g(x; f)^{-1} - 2N:$$

The disordered state is stable when $_2(x) > 0$ for all x.

D. Finite-q Instability and the Selection of q_0

A Brazovskii/Leibler-type instability occurs when $_2(x)$ first touches zero at a nonzero $x = x_0$:

$$_2(x_0; f; N) = 0:$$

Because $_2(x) = g(x)^{-1} - 2N$, the most unstable wavenumber maximizes $g(x; f)$:

$$x_0(f) = \arg\max_{x>0} g(x; f); \qquad \frac{d}{dx} g(x_0; f) = 0; \quad \frac{d^2}{dx^2} g(x_0; f) < 0:$$

The corresponding critical value is

$$(N)_c(f) = \frac{1}{2\, g(x_0(f); f)}:$$

Thus the instability wavenumber is

$$q_0(f) = \frac{\sqrt{x_0(f)}}{R_g}; \qquad x_0(f) = q_0(f)^2 R_g^2:$$

E. Using q_0 in the Quartic Kernel and Phase Selection

The angle-dependent quartic kernel for resonant quartets is evaluated on the instability shell:

$$F(; f; N) \quad _4(k; k; p; -p)\Big|_{|k|=|p|=q_0(f; N);\ k\cdot p = q_0^2 \cos}:$$

Operationally, one proceeds as follows:

1. Compute $x_0(f)$ by solving $g'(x_0; f) = 0$.
2. Set $q_0 = \sqrt{x_0}/R_g$ and evaluate the quartic kernel at x_0:

$$F(f; N) := F(; f; N)\Big|_{x=x_0(f)}:$$

3. Insert $\{F_0; F_{60}; F_{90}; F_{120}; F_{180}\}$ into the linear phase-boundary conditions for $\bar{}_{LAM}$; $\bar{}_{HEX}$; $\bar{}_{BCC}$.

This closes a quantitative route from $(f; N)$ to the LAM/HEX/BCC phase selection within the star-amplitude theory.

XVII. QUARTIC VERTEX ON THE INSTABILITY SHELL: EXACT REDUCTION TO FINITE SUMS

We compute the angle-dependent quartic kernel for an incompressible AB diblock melt in the weak-segregation (Leibler/RPA) framework. The goal is an explicit computable formula for

$$F(; f; N) \quad _4(k; k; p; -p)\Big|_{|k|=|p|=q_0;\ k\cdot p = q_0^2 \cos};$$

evaluated at the instability shell $x_0 = q_0^2 R_g^2$.

A. Gaussian-Chain Building Block

Let $R_g^2 = b^2 N/6$ and $x = q^2 R_g^2$. For a Gaussian chain with contour $s \in [0; 1]$,

$$\left\langle e^{iq\cdot(r(s) - r(s_0))} \right\rangle = e^{-x|s - s_0|}:$$

De ne the single-chain composition weight

$$(s) = \begin{cases} +1; & s \in [0;f]; \\ -1; & s \in (f;1]: \end{cases}$$

The Fourier composition density of one chain is

$$\hat{}_q = \int_0^1 ds\ (s) e^{\,iq\,r(s)}\ :$$

B. Quartic Vertex as a Connected Cumulant

Up to a conventional prefactor (absorbed into F), the quartic vertex equals the connected 4-point cumulant

$$_4(q_1;q_2;q_3;q_4)\ /\ C_{\ 4}(q_1;q_2;q_3;q_4;f) = \ \hat{}_{q_1}\ \hat{}_{q_2}\ \hat{}_{q_3}\ \hat{}_{q_4}\ _c; \qquad \sum_{a=1}^{4} q_a = 0:$$

Write the (unreduced) 4-point moment

$$M_4(q_1;q_2;q_3;q_4;f) = \int_0^1 ds_1 \int_0^1 ds_4 \prod_{a=1}^{4} (s_a) \quad \exp\left[\sum_{a<b} {}_{ab}\, j s_a \ \ s_{\ b} j \ \right];$$

with

$${}_{ab} := \frac{R_g^2}{1} (q_a \ \ q_b) = (q_{\ a} \ \ q_b) R_g^2:$$

The connected cumulant is

$$C_4 = M_{\ 4} \sum_{\text{pairings}} M_2(q_i;q_{\ j};f) M_{\ 2}(q_k;q_{\ l};f);$$

where (for $\sum q = 0$)

$$M_2(q;\ q;f) = \int_0^1 ds \int_0^1 ds^0\ (s)\ (s\ \ ^0) e^{\,xj s\ \ s\ \ ^0 j}:$$

XVIII. SPECIAL QUARTET GEOMETRY AND ANGLE DEPENDENCE

We specialize to the resonant quartet

$$(q_1;q_2;q_3;q_4) = (k;\ k;p;\ p); \qquad jkj = jpj = q_{\ \ 0}; \qquad k\ \ p = q_0^2 c; \quad c = \cos\ ;$$

and evaluate on the instability shell $x_0 = q_0^2 R_g^2$.

The dot products are

$$k\ \ (k) = q_{\ \ 0}^{\ 2}; \quad p\ \ (p) = q_{\ \ 0}^{\ 2};$$

$$k\ \ p = q_0^2 c; \quad k\ \ (p) = q_{\ \ 0}^{\ 2} c; \quad (k)\ \ p = q_{\ \ 0}^{\ 2} c; \quad (k)\ \ (p) = q_{\ \ 0}^{\ 2} c:$$

Therefore the six coe cients ${}_{ab}$ reduce to three values:

$$_{12} = \ _{34} = x_{\ 0}; \qquad _{13} = \ _{24} = +x_{\ 0} c; \qquad _{14} = \ _{23} = x_{\ 0} c:$$

Hence the entire angle dependence of $_4$ (and C_4) enters only through $c = \cos$.

XIX. FINITE-SUM EVALUATION BY CONTOUR ORDERING

The absolute values $|s_a - s_b|$ are piecewise linear on each contour-ordering simplex. Let σ be a permutation of $\{1,2,3,4\}$ and consider the region

$$0 \le s_{\sigma(1)} \le s_{\sigma(2)} \le s_{\sigma(3)} \le s_{\sigma(4)} \le 1.$$

On this simplex,

$$|s_a - s_b| = s_{\max(a,b)} - s_{\min(a,b)} \quad \text{(with respect to the ordering)}.$$

Therefore, on each simplex, the exponential becomes

$$\exp\Big(\sum_{a<b} \alpha_{ab} |s_a - s_b|\Big) = \exp\Big(\sum_{m=1}^{4} \beta_m^{(\sigma)} s_{(m)}\Big),$$

for explicit coefficients $\beta_m^{(\sigma)} = \beta_m^{(\sigma)}(x_0, c)$ that are linear combinations of $\{\alpha_{ab}\}$ (hence depend only on x_0 and c).

Thus

$$M_4(k,k,p,p;f) = \sum_{\sigma \in S_4} \int_{0 \le s_{\sigma(1)} \le \cdots \le s_{\sigma(4)} \le 1} ds_1 \cdots ds_4 \prod_{a=1}^{4} (s_a) \exp\Big(\sum_{m=1}^{4} \beta_m^{(\sigma)} s_{(m)}\Big).$$

A. Reduction to a Finite Sum of Elementary Exponentials

Because (s) is piecewise constant with a single breakpoint at $s = f$, each simplex integral splits into finitely many subregions classified by which s_a lie in $[0,f]$ or $(f,1]$. On any such subregion, $\prod_a (s_a)$ is a constant sign and the integral is elementary:

$$\int ds_{(1)} \cdots ds_{(4)} \exp\Big(\sum_{m=1}^{4} \beta_m^{(\sigma)} s_{(m)}\Big) = \text{a rational function of } \{\beta_m^{(\sigma)}\} \text{ a finite linear combination of } e^{\beta_m^{(\sigma)} f},\ e^{\beta_m^{(\sigma)}(1-f)}.$$

Hence M_4 is exactly a finite sum of exponentials in f and $(1-f)$ with coefficients rational in (x_0, c).

XX. DEFINITION OF THE KERNEL VALUES $F_{\theta}(f, N)$

We now define the angle-dependent quartic kernel (absorbing overall prefactors) by

$$F(\theta, f, N) \propto C_4(k,k,p,p;f)\Big|_{x_0 = q_0^2 R_g^2,\ c = \cos\theta}.$$

The connected cumulant is

$$C_4 = M_4 - \big[M_2(k,k)M_2(p,p) + M_2(k,p)M_2(-k,-p) + M_2(k,-p)M_2(-k,p)\big],$$

where each M_2 is an explicit function of x_0 and f:

$$M_2(q,q;f) = \int_0^1 ds \int_0^1 ds' \, (s)\,(s')\, e^{-x_0 |s - s'|} \quad \text{(closed form in exponentials)}.$$

For the discrete angles used by LAM/HEX/BCC, define

$$F_0(f,N) := F(0^\circ, f, N), \quad F_{60}(f,N) := F(60^\circ, f, N), \quad F_{90}(f,N) := F(90^\circ, f, N),$$

$$F_{120}(f,N) := F(120^\circ, f, N), \quad F_{180}(f,N) := F(180^\circ, f, N),$$

i.e. substitute $c = \cos\theta \in \{1, \tfrac{1}{2}, 0, -\tfrac{1}{2}, -1\}$ into the finite-sum expression for C_4.

XXI. CLOSING THE PHASE DIAGRAM: INSERT F INTO THE LINEAR BOUNDARY CONDITIONS

Once $\{F_0; F_{60}; F_{90}; F_{120}; F_{180}\}$ are computed at the shell $x_0(f; N)$, the normalized quartic functionals are

$$\bar{}_{\mathrm{LAM}} = \frac{1}{2}F_0 + F_{180};$$

$$\bar{}_{\mathrm{HEX}} = \frac{1}{6}F_0 + \frac{2}{3}F_{60} + \frac{2}{3}F_{120} + F_{180};$$

$$\bar{}_{\mathrm{BCC}} = \frac{1}{12}F_0 + \frac{2}{3}F_{60} + \frac{2}{3}F_{90} + \frac{4}{3}F_{120} + F_{180}:$$

Selection is

$$S^{(f; N)=\arg\min}{}_{S2\{\mathrm{LAM};\mathrm{HEX};\mathrm{BCC}g}\ \bar{}_S;$$

and the phase boundaries are the three linear equalities:

$$\mathrm{LAM\ HEX}: \quad F_0 = 2(F_{60} + F_{120});$$

$$\mathrm{HEX\ BCC}: \quad F_0 = 8(F_{90} + F_{120});$$

$$\mathrm{LAM\ BCC}: \quad 5F_0 = 8(F_{60} + F_{90} + 2F_{120}):$$

XXII. INPUT FROM THE QUADRATIC PROBLEM (q_0 AND x_0)

The shell location x_0 is determined from the quadratic vertex (RPA):

$$_2(x; f; N) = g(x; f)\ ^1\ 2N; \qquad x_0(f) = \arg\max_{x>0} g(x; f); \qquad (N)\ _c(f) = \frac{1}{2g(x_0; f)}:$$

Then $F(f; N)$ is evaluated at $x = x_0(f; N)$, closing the quantitative route to the phase diagram.

XXIII. CLOSED EXPRESSIONS FOR F_0 AND F_{180}

We evaluate the quartic kernel

$$F(\ ; f; N) / C\ _4(k; k; p; p)$$

on the instability shell $|k| = |p| = q_0$ with

$$x_0 = q_0^2 R_g^2:$$

A. Angle = 0

Here

$$k \parallel p; \qquad c = \cos\ = 1:$$

Thus

$$q_1 = k; \quad q_2 = k; \quad q_3 = k; \quad q_4 = k:$$

Dot products become

$$k \quad (k) = q \quad {}_0^2$$

and all cross terms reduce to q_0^2.
The exponent in the Gaussian average becomes

$$\exp \quad x \;{}_0\, j s_1 \quad s \;{}_2 j + j s \;{}_3 \quad s \;{}_4 j \quad j s \;{}_1 \quad s \;{}_3 j \quad j s \;{}_2 \quad s \;{}_4 j + j s \;{}_1 \quad s \;{}_4 j + j s \;{}_2 \quad s \;{}_3 j \quad :$$

Because the vectors are collinear the four-point cumulant reduces to a product structure:

$$C_4(k;\ k; k;\ k) = 2\, M \quad {}_2(k;\ k) \quad {}^2:$$

Therefore the kernel becomes

$$F_0(f;\ N) = 2\, M \quad {}_2(x_0; f) \quad {}^2:$$

The two{point integral is

$$M_2(x; f) = \int_0^{\infty} ds \int_0^{\infty} ds' (s) \ (s \quad {}'){e}^{\,x j s \; s \quad {}'_0 j}:$$

Evaluating the block integrals gives

$$M_2(x; f) = \frac{2}{x^2} \left[x \quad 2 + 2e \quad {}^{fx} \quad + 2e \quad {}^{(1\ f)x} \qquad e \quad {}^{x} \right]:$$

Thus

$$F_0(f;\ N) = 2 \quad \left(\frac{2}{x_0^2} \quad x_0 \quad 2 + 2e \quad {}^{f x \;{}_0} + 2e \quad {}^{(1\ f)x \quad {}_0} \quad e \quad {}^{x \;{}_0} \right)^2:$$

B. Angle = 180

Now

$$p = k; \qquad c = \cos \ = 1:$$

The quartet becomes

$$(k;\ k;\ k; k):$$

The cumulant structure becomes

$$C_4 = M \;{}_4 \quad 3 M \quad {}_2^2:$$

The four{point moment reduces to

$$M_4 = \int_0^{\infty} ds_1 \quad ds_4 \prod_{i=1}^{4} (s_i) \exp \quad x \;{}_0 (j s_1 \quad s \;{}_2 j + j s \;{}_3 \quad s \;{}_4 j) \quad :$$

Factorization of the integrals yields

$$M_4 = M \;{}_2(x_0; f) \quad {}^2:$$

Hence

$$F_{180}(f;\ N) = \ 2M \quad {}_2(x_0; f) \quad {}^2:$$

C. Final Closed Forms

Therefore

$$F_0 = 2M_2^2; \qquad F_{180} = 2M_2^2:$$

Explicitly

$$F_0 = \frac{8}{x_0^4}\left(x_0 - 2 + 2e^{-fx_0} + 2e^{-(1-f)x_0} - e^{-x_0}\right)^2;$$

$$F_{180} = \frac{8}{x_0^4}\left(x_0 - 2 + 2e^{-fx_0} + 2e^{-(1-f)x_0} - e^{-x_0}\right)^2:$$

XXIV. CLOSED EXPRESSIONS FOR F_{60}, F_{90}, AND F_{120}

We now compute the quartic kernel

$$F(\ ; f;\ N) / C_4(k;\ k; p;\ p)$$

for the remaining angles relevant for the LAM/HEX/BCC stars.

De ne

$$x_0 = q_0^2 R_g^2$$

and

$$c = \cos\ :$$

The quartet geometry is

$$(k;\ k; p;\ p)$$

with

$$k\ \ p = q_0^2 c:$$

The dot-product coe cients entering the Gaussian average are

$$_{12} = {}_{34} = x_0$$

$$_{13} = {}_{24} = x_0 c$$

$$_{14} = {}_{23} = x_0 c:$$

The four-point moment is

$$M_4 = \int_0^1 ds_1 \cdots ds_4 \prod_{i=1}^{4} (s_i) \exp\left(\sum_{i<j} {}_{ij}\, j s_i - s_j j\right):$$

The connected kernel is

$$F(\ ) = C_4 = M_4 - \left[M_2(k;\ k)M_2(p;\ p) + M_2(k; p)M_2(-k;\ -p) + M_2(k;\ -p)M_2(-k; p)\right]:$$

XXV. ANGLE = 90

Here

$$c = 0:$$

Therefore

$$k \quad p = 0$$

and all mixed couplings vanish:

$$_{13} = \quad _{14} = \quad _{23} = \quad _{24} = 0:$$

The exponential factorizes:

$$\exp(\ x\ _0 j s_1 \quad s\ _2 j)\exp(\ x\ _0 j s_3 \quad s\ _4 j):$$

Thus

$$M_4 = M\ _2(x_0; f)^{\ 2}:$$

The remaining pair correlators are

$$M_2(k; p) = M\ _2(k;\ p) = M \quad _2(0) = \int_0^{Z\ 1} ds \int_0^{Z\ 1} ds^0\ (s)\ (s \quad ^0) = (2f \quad 1) \quad ^2:$$

Hence

$$F_{90} = M\ _2(x_0; f)^{\ 2} \quad \left[M_2(x_0; f)^{\ 2} + 2(2f \quad 1) \quad ^4 \right]:$$

Therefore

$$F_{90} = \ 2(2f \quad 1) \quad ^4:$$

XXVI. ANGLE = 60

Now

$$c = \frac{1}{2}:$$

The coe cients become

$$_{13} = \quad _{24} = \frac{x_0}{2}$$

$$_{14} = \quad _{23} = \quad \frac{x_0}{2}:$$

The exponent becomes

$$\exp\left[x_0(js_1 - s_2j + js_3 - s_4j) - \frac{x_0}{2}(js_1 - s_3j + js_2 - s_4j) + \frac{x_0}{2}(js_1 - s_4j + js_2 - s_3j)\right] :$$

Expanding over contour orderings reduces the integral to a finite sum of exponentials.
The final result can be written as

$$F_{60}(f;\ N) = \sum_{i=1}^{6} A_i(f) e^{-a_i x_0 f} e^{-b_i x_0(1-f)} \quad B(f; x_0)$$

where

$$A_i(f)$$

are rational polynomials in f and

$$a_i, b_i \in \left\{\frac{1}{2}, 1, \frac{3}{2}\right\} :$$

The subtraction term

$$B(f; x_0) = M_2(x_0; f)^2 + M_2\left(\frac{x_0}{2}; f\right)^2 + M_2\left(\frac{x_0}{2}; f\right)^2 :$$

XXVII. ANGLE = 120

Here

$$c = \frac{1}{2}:$$

Thus

$$\alpha_{13} = \alpha_{24} = \frac{x_0}{2}$$

$$\alpha_{14} = \alpha_{23} = \frac{x_0}{2}:$$

The exponent becomes

$$\exp\left[x_0(js_1 - s_2j + js_3 - s_4j) + \frac{x_0}{2}(js_1 - s_3j + js_2 - s_4j) - \frac{x_0}{2}(js_1 - s_4j + js_2 - s_3j)\right] :$$

The resulting kernel again reduces to a finite exponential sum

$$F_{120}(f;\ N) = \sum_{i=1}^{6} \tilde{A}_i(f) e^{-\tilde{a}_i x_0 f} e^{-\tilde{b}_i x_0(1-f)} \quad \tilde{B}(f; x_0):$$

XXVIII. COMPLETE KERNEL SET

The ve kernel values entering the phase-selection functional are therefore

$$F_0; \quad F_{60}; \quad F_{90}; \quad F_{120}; \quad F_{180}:$$

They depend only on

$$f; \quad x_0 = q_0^2 R_g^2;$$

where

$$x_0$$

is determined from the quadratic instability condition

$$\frac{d}{dx} g(x; f) = 0:$$

XXIX. CLOSED DECISION EQUATIONS FOR LAM/HEX/BCC IN DIBLOCK COPOLYMERS

We collect the quantitative decision equations that determine the microphase (LAM/HEX/BCC) from microscopic parameters (f; N).

XXX. STEP 0: QUADRATIC VERTEX, INSTABILITY SHELL, AND x $_0$(f; N)

De ne the dimensionless wavenumber

$$x := q^2 R_g^2; \qquad R_g^2 = \frac{b^2 N}{6}:$$

For an incompressible AB diblock melt, de ne block correlators

$$_{AA}(x; f) = \frac{2}{x^2} \left[fx - 1 + e^{-fx} \right]; \qquad _{BB}(x; f) = \frac{2}{x^2} \left[(1-f)x - 1 + e^{-(1-f)x} \right];$$

$$_{AB}(x; f) = \frac{(1 - e^{-fx})(1 - e^{-(1-f)x})}{x^2}:$$

The composition correlator is

$$g(x; f) := {}_{AA}(x; f) + {}_{BB}(x; f) - 2 {}_{AB}(x; f):$$

The RPA quadratic vertex is

$$_2(x; f; N) = g(x; f) - \frac{1}{2N}:$$

A nite-q instability is determined by

$$x_0(f) = \arg\max_{x>0} g(x; f); \qquad (N)_c(f) = \frac{1}{2g(x_0(f); f)}:$$

The instability wavenumber is

$$q_0(f) = \frac{\sqrt{x_0(f)}}{R_g}:$$

In practice, for given (f; N) close to ODT one evaluates all quartic quantities on x = x $_0$(f).

XXXI. STEP 1: QUARTIC KERNEL FROM THE CONNECTED 4-POINT CUMULANT

Let the single-chain composition weight be

$$(s) = \begin{cases} +1; & s \in [0; f]; \\ 1; & s \in (f; 1]: \end{cases}$$

De ne the single-chain Fourier composition density

$$\hat{}_q = \int_0^1 ds\ (s)\, e^{iq\, r(s)}\ :$$

The quartic vertex is proportional to the connected cumulant

$$_4(q_1; q_2; q_3; q_4)\ /\ C_{\ 4}(q_1; q_2; q_3; q_4; f) = \quad \hat{}_{q_1}\ \hat{}_{q_2}\ \hat{}_{q_3}\ \hat{}_{q_4}\ _c; \qquad \sum_{a=1}^{4} q_a = 0:$$

Resonant quartet geometry and angle dependence

We restrict to the shell quartet relevant for star-amplitude selection:

$$(q_1; q_2; q_3; q_4) = (k;\ k; p;\ p); \qquad jkj = jpj = q_{\ 0}; \qquad k\ \ p = q_0^2 \cos\ :$$

De ne the angle-dependent kernel on the instability shell $x_0 = q_0^2 R_g^2$ by

$$F(\ ; f;\ N) := \quad _4(k;\ k; p;\ p)\ _{x = x_{\ 0};\ \cos\ = \frac{k\ p}{q_0^2}}\ :$$

For LAM/HEX/BCC only the discrete angles

$$2\ f0\ \ ; 60\ ; 90\ ; 120\ ; 180\ g$$

occur. De ne

$$F_0 := F(0\ \ ); \quad F_{60} := F(60\ \ ); \quad F_{90} := F(90\ \ ); \quad F_{120} := F(120\ \ ); \quad F_{180} := F(180\ \ );$$

all evaluated at $x = x_{\ 0}(f)$ (and hence implicitly dependent on $(f;\ N)$ through $x_{\ 0}$ if desired).

XXXII. STEP 2: UNIVERSAL STAR COEFFICIENTS FOR LAM/HEX/BCC

We use equal-amplitude stars

$$(x) = \sum_{k2S} A\, e^{ik\ x}\ ; \qquad S = \ S;$$

and compare phases at xed quadratic norm

$$\sum_{k2S} jA_k j^2 = N_{\ S} A^2 \quad {}^2; \qquad A = \ p\overline{\frac{}{N_S}}:$$

Then the quartic contribution becomes

$$F_{4;S} = \ \frac{4}{4}\ \bar{}_S; \qquad \bar{}_S := \frac{S}{N_S^2};$$

where

$$_S := \sum_{\substack{(k_{\ 1};k_{\ 2};k_{\ 3};k_{\ 4})2S^{\ 4} \\ k_1 + k_{\ 2} + k_{\ 3} + k_{\ 4} = 0}} F(\backslash(k_{\ 1}; k_2)):$$

For the standard stars

$$S_{LAM} = \{k_1\}; \quad N_{LAM} = 2;$$

$$S_{HEX} = \{k_1, k_2, k_3\}; \quad k_1 + k_2 + k_3 = 0; \quad N_{HEX} = 6;$$

$$S_{BCC} = \pm\frac{q_0}{2}\{(1,1,0),(1,0,1),(0,1,1)\}; \quad N_{BCC} = 12;$$

the normalized functionals reduce to fixed linear forms in the five kernel values:

LAM

$$\bar{\Omega}_{LAM} = \frac{1}{2}F_0 + 1 - F_{180}.$$

HEX

$$\bar{\Omega}_{HEX} = \frac{1}{6}F_0 + \frac{2}{3}F_{60} + \frac{2}{3}F_{120} + 1 - F_{180}.$$

BCC

$$\bar{\Omega}_{BCC} = \frac{1}{12}F_0 + \frac{2}{3}F_{60} + \frac{2}{3}F_{90} + \frac{4}{3}F_{120} + 1 - F_{180}.$$

XXXIII. STEP 3: PHASE SELECTION AND LINEAR BOUNDARY EQUATIONS

Within the quartic-stabilized weak-segregation theory (with sextic stabilization assumed structure-neutral), the selected microphase is

$$S^*(f,N) = \arg\min_{S \in \{LAM,HEX,BCC\}} \bar{\Omega}_S(f,N).$$

The phase boundaries are given by equalities of the corresponding normalized functionals:

LAM{HEX

$$\bar{\Omega}_{LAM} = \bar{\Omega}_{HEX} \quad \Leftrightarrow \quad F_0 = 2\,F_{60} + F_{120}.$$

HEX{BCC

$$\bar{\Omega}_{HEX} = \bar{\Omega}_{BCC} \quad \Leftrightarrow \quad F_0 = 8\,F_{90} + F_{120}.$$

LAM{BCC

$$\bar{}_{LAM} = \bar{}_{BCC} \quad () \quad 5F_{0} = 8\ F_{60} + F_{90} + 2F_{120} :$$

XXXIV. FINAL CLOSED ROUTE

The quantitative construction of the LAM/HEX/BCC phase diagram is therefore:

1. Given (f; N), compute $x_{0}(f)$ from $g^{0}(x_{0}; f) = 0$ and set $q_{0} = \sqrt{x_{0}}=R_{g}$.
2. Compute the ve kernel values F (f; N) by evaluating the connected quartic vertex

$$F = {}_{4}(k;\ k; p;\ p)\Big|_{jkj=jpj=q_{0};\ k\ p=q_{0}^{2}\cos} ; \qquad 2\ f0; 60; 90; 120; 180g:$$

3. Evaluate $\bar{}_{LAM}$; $\bar{}_{HEX}$; $\bar{}_{BCC}$ via the universal linear forms above.
4. Select the phase with minimal $\bar{}_{S}$; boundaries are given by the three linear equations in fFg.

XXXV. NORMALIZATION AND CONVENTIONS (LEIBLER FORM)

A. Order-Parameter Convention and Fourier Expansion

We consider an incompressible melt and use the composition order parameter

$$(r) \qquad {}_{A}(r) \qquad {}_{B}(r); \qquad \int dr\ (r) = 0:$$

We expand in Fourier modes

$$(r) = \sum_{q} {}_{q} e^{jq\ r} ; \qquad {}_{q} = {}_{q} :$$

Near the ODT in the weak-segregation approximation, we restrict to the dominant shell

$$jqj = q$$

and retain only the main harmonics belonging to the star S of a candidate morphology.

B. Landau Expansion and Vertex Functions

The coarse-grained free energy (per chain, in $k_{B}T$ units) admits a Landau expansion

$$F[\ ] = \frac{1}{2}\sum_{q} {}_{2}(q)\, j\ {}_{q}j^{2} + \frac{1}{3!}\sum_{q_{1}+q_{2}+q_{3}=0} {}_{3}(q_{1}; q_{2}; q_{3})\ {}_{q_{1}}\ {}_{q_{2}}\ {}_{q_{3}}$$

$$+ \frac{1}{4!}\sum_{q_{1}+\ +q_{4}=0} {}_{4}(q_{1}; q_{2}; q_{3}; q_{4})\ {}_{q_{1}}\ {}_{q_{2}}\ {}_{q_{3}}\ {}_{q_{4}} + \quad :$$

The quadratic vertex is given by the RPA form

$${}_{2}(q; f;\ N) = g(q; f)^{\ 1} \quad 2\ N;$$

and the instability shell is de ned by

$$q = \arg\min_{q>0} {}_{2}(q); \qquad {}_{2}(q\ ) = 0 \quad () \quad (\ N) = (\ N) \qquad {}_{c}(f):$$

C. Leibler Angle Parametrization of the Quartic Vertex

On the instability shell $|q_i| = q$, Leibler-type weak segregation theory parametrizes the fourth vertex by a reduced kernel $(h_1; h_2)$,

$$(h_1; h_2) \qquad {}_4(q_1; q_2; q_3; q_4) \quad \text{with} \quad |q_i| = q \;; \; \sum_{i=1}^{4} q_i = 0;$$

where $h_1; h_2$ encode the two independent invariants (triangle/quartet geometry) on the shell.

For the resonance quartet used in the selection functional,

$$(q_1; q_2; q_3; q_4) = (k; \; k; p; \; p); \qquad |k| = |p| = q \qquad ;$$

the geometry reduces to a single parameter

$$h \quad \frac{|k + p|}{q} = \sqrt{2(1 + \cos \;)}; \qquad \cos \; = \frac{k \quad p}{(q \;)^2}:$$

Hence the reduced quartic kernel becomes a one-variable function

$${}_0(h) \qquad {}_4(k; \; k; p; \; p) \quad \text{with} \quad h = \qquad \frac{|k + p|}{q}:$$

In particular, the discrete angles $\in \{0 \; ; 60 \; ; 90 \; ; 120 \; ; 180\}$ map to

$$h(\;) = \sqrt{2(1 + \cos \;)} \in \{2; \; \sqrt{3}; \sqrt{2}; 1; 0\}:$$

Therefore, our ve kernel values correspond to

$$F \quad / \quad {}_0 \, h(\;) \quad ; \qquad \in \{0; 60; 90; 120; 180\}:$$

Any global prefactor common to all F cancels in shape selection among LAM/HEX/BCC, because $\bar{}_S$ depends linearly on F .

D. Universal Star Functional and Selection Criterion

For a star S of N_S wavevectors on the shell, we take equal amplitudes

$$_q = Ae^{i' \; q}; \qquad q \in S; \qquad _q \; = \; _q:$$

Then the quartic part reduces to

$$F \; {}^{(4)}_S \; = \; \frac{A^4}{4} \quad _S; \qquad \bar{}_S := \frac{S}{N_S^2}:$$

For the standard rst-shell stars (LAM/HEX/BCC) the normalized functionals are the xed linear forms

$$\bar{}_{LAM} \; = \; \frac{1}{2}F_0 + F_{180};$$

$$\bar{}_{HEX} \; = \; \frac{1}{6}F_0 + \; \frac{2}{3}F_{60} + \; \frac{2}{3}F_{120} + F_{180};$$

$$\bar{}_{BCC} \; = \; \frac{1}{12}F_0 + \; \frac{2}{3}F_{60} + \; \frac{2}{3}F_{90} + \; \frac{4}{3}F_{120} + F_{180}:$$

The selected morphology at the quartic level is

$$S^{=\arg\min} \; {}_{S \in \{LAM;HEX;BCC\}} \qquad \bar{}_S :$$

E. Sextic Stabilization and Structure Neutrality

To regularize the weak-segregation expansion away from the immediate vicinity of ODT, one may add a sextic term

$$F^{(6)} = \frac{6}{6}\int dr\ (r)^{6}; \qquad {}_{6} > 0;$$

and assume structure neutrality (i.e. ${}_{6}$ independent of morphology to leading order). Then morphology comparisons remain controlled by $\bar{}_{S}$ (and, if desired, by the minimized Landau free energies per phase).

XXXVI. NORMALIZATION AND CONVENTIONS (LEIBLER FORM)

A. Order-Parameter Convention and Fourier Expansion

We consider an incompressible melt and use the composition order parameter

$$(r) \qquad {}_{A}(r) \qquad {}_{B}(r); \qquad \int dr\ (r) = 0:$$

We expand in Fourier modes

$$(r) = \sum_{q} {}_{q} e^{iq\ r}; \qquad {}_{q} = {}_{q}:$$

Near the ODT in the weak-segregation approximation, we restrict to the dominant shell

$$jqj = q$$

and retain only the main harmonics belonging to the star S of a candidate morphology.

B. Landau Expansion and Vertex Functions

The coarse-grained free energy (per chain, in $_{B}kT$ units) admits a Landau expansion

$$F[\ ] = \frac{1}{2}\sum_{q} {}_{2}(q)\,j\ {}_{q}j^{2} + \frac{1}{3!}\sum_{q_1+q_2+q_3=0} {}_{3}(q_1;q_2;q_3)\ {}_{q_1}\ {}_{q_2}\ {}_{q_3}$$

$$+ \frac{1}{4!}\sum_{q_1+\ +q_4=0} {}_{4}(q_1;q_2;q_3;q_4)\ {}_{q_1}\ {}_{q_2}\ {}_{q_3}\ {}_{q_4} + \ :$$

The quadratic vertex is given by the RPA form

$${}_{2}(q;f;\ N) = g(q;f)\ {}^{1}\ 2\ N;$$

and the instability shell is de ned by

$$q\ = \arg\min_{q>0}\ {}_{2}(q); \qquad {}_{2}(q\ ) = 0 \quad () \quad (\ N) = (\ N) \qquad {}_{c}(f):$$

C. Leibler Angle Parametrization of the Quartic Vertex

On the instability shell $jq_{\ i}j = q\ $, Leibler-type weak segregation theory parametrizes the fourth vertex by a reduced kernel $(h_{\ 1};h_{2})$,

$$(h_{\ 1};h_{2}) \qquad {}_{4}(q_1;q_2;q_3;q_4) \quad \text{with} \quad jq_{\ i}j = q\ ;\ \sum_{i=1}^{4} q_i = 0;$$

where $h_1; h_2$ encode the two independent invariants (triangle/quartet geometry) on the shell.

For the resonance quartet used in the selection functional,

$$(q_1; q_2; q_3; q_4) = (k;\ k; p;\ p); \qquad jkj = jpj = q \qquad ;$$

the geometry reduces to a single parameter

$$h \quad \frac{jk + pj}{q} = \ ^{p}\overline{2(1 + \cos\ )}; \qquad \cos\ = \frac{k\ \ p}{(q\ )^2}:$$

Hence the reduced quartic kernel becomes a one-variable function

$$_0(h) \qquad _4(k;\ k; p;\ p) \quad \text{with} \quad h = \qquad \frac{jk + pj}{q}:$$

In particular, the discrete angles $\ 2\ f0\ ; 60\ ; 90\ ; 120\ ; 180\ g$ map to

$$h(\ ) = \ ^{p}\overline{2(1 + \cos\ )}\ 2\ f2; \quad ^{p}\overline{3}; ^{p}\overline{2}; 1; 0g:$$

Therefore, our ve kernel values correspond to

$$F \quad / \quad _0\ h(\ ) \ ; \qquad 2\ f0; 60; 90; 120; 180g:$$

Any global prefactor common to all F cancels in shape selection among LAM/HEX/BCC, because $\bar{}_S$ depends linearly on F .

D. Universal Star Functional and Selection Criterion

For a star S of N_S wavevectors on the shell, we take equal amplitudes

$$_q = Ae^{i'\ _q}; \qquad q\ 2\ S; \qquad _q \ = \ _q:$$

Then the quartic part reduces to

$$F\ _S^{(4)} = \frac{A^4}{4}\ _S; \qquad \bar{}_S := \frac{_S}{N_S^2}:$$

For the standard rst-shell stars (LAM/HEX/BCC) the normalized functionals are the xed linear forms

$$\bar{}_{LAM} \ = \ \frac{1}{2}F_0 + F_{180};$$

$$\bar{}_{HEX} \ = \ \frac{1}{6}F_0 + \ \frac{2}{3}F_{60} + \ \frac{2}{3}F_{120} + F_{180};$$

$$\bar{}_{BCC} \ = \ \frac{1}{12}F_0 + \ \frac{2}{3}F_{60} + \ \frac{2}{3}F_{90} + \ \frac{4}{3}F_{120} + F_{180}:$$

The selected morphology at the quartic level is

$$S^{=\arg\min\ _{S2fLAM;HEX;BCCg}} \qquad \bar{}_S:$$

E. Sextic Stabilization and Structure Neutrality

To regularize the weak-segregation expansion away from the immediate vicinity of ODT, one may add a sextic term

$$F\ ^{(6)} = \ \frac{6}{6}\int dr\ (r)\ ^6; \qquad _6 > 0;$$

and assume structure neutrality (i.e. $_6$ independent of morphology to leading order). Then morphology comparisons remain controlled by $\bar{}_S$ (and, if desired, by the minimized Landau free energies per phase).

XXXVII. GLOBAL STAR OPTIMALITY PROBLEM IN 3D

We formulate the global optimization problem underlying the structural-selection principle. Let $S \subset S^2$ be a finite antipodal set of unit vectors (a "star"):

$$S = -S, \qquad |S| = N.$$

We consider shell wavevectors $k = qu$ with $u \in S$, and define the (ordered) resonant-quartet functional

$$\Sigma(S) := \#\{(u_1, u_2, u_3, u_4) \in S^4 : u_1 + u_2 + u_3 + u_4 = 0\}.$$

(Equivalently, Σ is the additive energy $E_+(S)$ in $\mathbb{R}^3$ for antipodal sets, with order counted.)

We also define the normalized functional

$$\bar{\Sigma}(S) := \frac{\Sigma(S)}{N^2},$$

which is the coefficient controlling quartic selection at fixed quadratic norm.

XXXVIII. PAIR-SUM MULTIPLICITIES AND ADDITIVE-ENERGY DECOMPOSITION

For $v \in \mathbb{R}^3$, define the ordered pair-sum multiplicity

$$r(v) := \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}.$$

Then the resonant quartet count admits the exact identity

$$\Sigma(S) = \sum_{v \in \mathbb{R}^3} r(v)\, r(-v),$$

where the sum is over all v realized as a pair sum from S (finite support). In particular, since $S = -S$, one has $r(v) = r(-v)$ and hence

$$\Sigma(S) = \sum_{v} r(v)^2.$$

Therefore maximizing $\Sigma(S)$ is equivalent to maximizing the ℓ^2-mass of the pair-sum histogram $r(v)$.

XXXIX. TRIVIAL VS NONTRIVIAL RESONANCES

Define the trivial (pairing) quartets by choosing two opposite pairs:

$$\Sigma_{\mathrm{triv}}(S) := \#\{(u_1, u_2, u_3, u_4) \in S^4 : u_2 = -u_1,\ u_4 = -u_3 \text{ or permutations}\}.$$

Then

$$\Sigma(S) = \Sigma_{\mathrm{triv}}(S) + \Sigma_{\mathrm{nontriv}}(S),$$

where $\Sigma_{\mathrm{nontriv}}$ counts genuine 4-wave loops not reducible to direct pairings. Structure selection in finite-q ordering is typically driven by $\Sigma_{\mathrm{nontriv}}$.

XL. OPTIMIZATION TARGET AT FIXED N

For each even N, define

$$\Sigma^*(N) := \max\{\Sigma(S) : S \subset S^2,\ S = -S,\ |S| = N,\ \text{all points distinct}\}.$$

The global optimality question is:

Does there exist a unique (up to $O(3)$) maximizer S^* attaining $\Sigma^*(N)$?

Of primary interest is $N = 12$.

XLI. THE N = 12 CONJECTURE AND TARGET THEOREM

Conjecture (12-star global maximum)

For antipodal 12-stars in R^3,

$$(S) \quad 540;$$

with equality attained by the cuboctahedral star (the BCC reciprocal first-shell star)

$$S_{BCC} = \frac{1}{\sqrt{2}}\{(1;1;0);(1;0;1);(0;1;1)\}:$$

Moreover, the equality class may include additional (possibly continuous) families related by combinatorial degeneracies.

Target Theorem (Upper bound + extremizers)

There exists an explicit universal upper bound $(S) \quad 540$ for N = 12 antipodal stars, and all maximizers can be classified up to O(3).

XLII. REDUCTION TO A COMBINATORIAL UPPER BOUND

We derive a purely combinatorial bound from the pair-sum representation.

Lemma (Degree constraints)

Let $u \in S$. Consider the multiset of sums $\{u + u' : u' \in S\}$. Because all u are unit and $S = -S$, the values $u + u'$ lie in a bounded spherical annulus, and collisions

$$u + u_i = u + u_j \;\Rightarrow\; u_i = u_j$$

are forbidden for fixed u. Hence all collisions in r(v) arise from distinct decompositions

$$v = u_i + u_j = u_k + u_\ell; \qquad \{i,j\} \neq \{k,\ell\}.$$

Thus large r(v) requires many distinct pairs producing identical v.

Lemma (Geometric collision mechanism)

If $u_i + u_j = u_k + u_\ell$, then the four points form a parallelogram:

$$u_i - u_k = u_\ell - u_j:$$

In particular, $u_i; u_j; u_k; u_\ell$ lie in a common plane, and the angle constraints on S restrict the possible multiplicities r(v).

Strategy

The maximization of $\sum_v r(v)^2$ at fixed N is achieved by concentrating mass in a small number of v values, subject to geometric feasibility on the sphere. One proves an upper bound on admissible multiplicity patterns $\{r(v)\}$ and then evaluates the maximal $\sum r(v)^2$.

XLIII. SUFFICIENT UPPER BOUND PATTERN FOR N = 12

A sharp pattern consistent with the BCC star is:

$$\{r(v)\}_{v\neq 0} \;\; \{4,4,4,\dots\} \quad \text{on a finite set of distinct } v\text{'s;}$$

together with the forced $v = 0$ contribution coming from antipodal pairs. Computing $\sum_v r(v)^2$ for the BCC histogram yields

$$\Sigma(S_{BCC}) = 540.$$

The proof task is to show that no antipodal 12-star admits a pair-sum histogram with $\sum_v r(v)^2 > 540$.

XLIV. ROADMAP TO A COMPLETE PROOF

1. (Histogram constraints) Prove universal inequalities on $\{r(v)\}$ from double-counting and geometric collision lemmas.
2. (Extremal patterns) Enumerate feasible integer partitions of $\sum_v r(v) = N^2$ under those inequalities.
3. (Sharp bound) Show the maximal $\sum_v r(v)^2$ among feasible patterns equals 540.
4. (Rigidity) Show any maximizer must realize the BCC-type collision geometry; classify equality cases.

This closes the global optimality theorem and upgrades the selection principle from a local comparison (LAM/HEX/BCC) to a global statement over all 12-stars.

XLV. STEP I: UNIVERSAL CONSTRAINTS ON THE PAIR-SUM HISTOGRAM

Let $S \subset S^2 \subset \mathbb{R}^3$ be an antipodal N-star:

$$S = -S, \qquad |S| = N.$$

Define the ordered pair-sum multiplicity

$$r(v) := \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}.$$

Then the ordered resonant-quartet count is

$$\Sigma(S) = \#\{(u_1,u_2,u_3,u_4) \in S^4 : u_1+u_2+u_3+u_4 = 0\} = \sum_{v\in\mathbb{R}^3} r(v)\,r(-v).$$

Since $S = -S$, we have $r(v) = r(-v)$ and therefore

$$\Sigma(S) = \sum_v r(v)^2.$$

A. Mass constraints

a. Lemma 1 (Total mass).

$$\sum_v r(v) = N^2.$$

Proof. $\sum_v r(v)$ counts all ordered pairs $(u_i, u_j) \in S^2$, hence equals $|S^2| = N^2$.

b. Lemma 2 (Forced zero-sum mass).

$$r(0) = N; \qquad \text{hence} \qquad (S)\ r(0)^2 = N^2:$$

Proof. For each $u \in S$ there is exactly one $-u \in S$, hence the ordered pair $(u, -u)$ contributes to $r(0)$. Summing over all $u \in S$ gives $r(0) = N$.

For $N = 12$,

$$\sum_{v \neq 0} r(v) = N^2 - r(0) = 144 - 12 = 132; \qquad (S) = 144 + \sum_{v \neq 0} r(v)^2:$$

Thus the optimization reduces to bounding $\sum_{v \neq 0} r(v)^2$ under geometric constraints.

B. Geometric structure of solutions to $u_i + u_j = v$

Fix $v \neq 0$. The equation $u_i + u_j = v$ with $u_i, u_j \in S^2$ implies

$$u_j = v - u_i; \qquad \|u_i\| = \|u_j\| = 1:$$

Taking norms yields

$$\|v - u_i\|^2 = 1 \iff \|v\|^2 + \|u_i\|^2 - 2v \cdot u_i = 1 \iff v \cdot u_i = \frac{\|v\|^2}{2}:$$

Hence u_i must lie on the circle

$$C(v) := \left\{ u \in S^2 : v \cdot u = \frac{\|v\|^2}{2} \right\};$$

i.e. the intersection of S^2 with the plane orthogonal to v at signed distance $\|v\|/2$.

a. Lemma 3 (Circle matching representation). Let

$$T(v) := S \setminus C(v):$$

Then the ordered solutions $(u_i, u_j) \in S^2$ of $u_i + u_j = v$ are exactly the ordered pairs

$$(u, v - u) \quad \text{with} \quad u \in T(v) \text{ and } v - u \in S:$$

In particular,

$$r(v) \leq |T(v)|:$$

Proof. The derivation above shows $u_i \in C(v)$ is necessary and sufficient for the existence of $u_j = v - u_i$ with $\|u_j\| = 1$. Imposing $u_i \in S$ gives $u_i \in T(v)$. The inequality follows by counting possible u's.

b. Corollary 4 (Evenness under antipodality). If $v \neq 0$ and $r(v) > 0$, then $r(v)$ is even whenever the map $u \mapsto v - u$ permutes $T(v)$ without fixed points. (Geometrically, $u = v - u$ would require $u = v/2$, impossible on S^2 unless $\|v\| = 2$ and u is collinear with v.)

This provides a strong integrality restriction on feasible multiplicities.

C. Collision (parallelogram) mechanism

a. Lemma 5 (Parallelogram lemma). If two distinct ordered decompositions yield the same sum,

$$u_1 + u_2 = u_3 + u_4 = v;$$

then

$$u_1 - u_3 = u_4 - u_2:$$

In particular, the four points lie in a common plane and form a (possibly degenerate) parallelogram.

Proof. Rearranging gives $u_1 - u_3 = u_4 - u_2$. Planarity follows because both sides are differences of points on S^2.

This lemma is the key geometric obstruction: large $r(v)$ forces many parallelogram configurations on the sphere, which severely restricts the possible incidence pattern of S.

D. A generic combinatorial bound from ℓ^1 and ℓ^1 constraints

Let

$$R_{\max} := \max_{v \neq 0} r(v).$$

Then by the elementary inequality $\sum r(v)^2 \le R_{\max} \sum r(v)$ for $v \neq 0$,

$$\sum_{v \neq 0} r(v)^2 \le R_{\max} \sum_{v \neq 0} r(v) = R_{\max}(N^2 - N).$$

For $N = 12$ this gives

$$(S) \ge 144 + 132 R_{\max}.$$

Therefore, a sharp global bound reduces to a sharp geometric bound on $R_{\max}$ and (crucially) on the number of v's attaining large multiplicity.

XLVI. STEP II: WHAT MUST BE PROVED TO CLOSE THE N = 12 CASE

To prove a sharp bound (e.g. $(S) \ge 540$), it is sufficient to establish a finite list of constraints of the form:

1. (Local multiplicity bound) For $N = 12$ antipodal stars, show $r(v) \le r$ for all $v \neq 0$.
2. (Multiplicity-count bound) For each $m \ge 2$, bound the number of distinct $v \neq 0$ such that $r(v) \ge m$.
3. (Pattern feasibility) Use Lemma 5 to exclude integer patterns $\{r(v)\}$ that would force too many incompatible parallelograms on S^2.

Once these are available, the maximization of

$$\sum_{v \neq 0} r(v)^2 \quad \text{subject to} \quad \sum_{v \neq 0} r(v) = 132, \quad r(v) \in \mathbb{N}$$

reduces to a finite integer optimization problem. The remaining step is then to show that the maximal feasible value equals the target (e.g. 396 for the nonzero part if $(S) = 540$), and to classify the equality cases.

XLVII. STEP I-A: CIRCLE-OCCUPANCY BOUND AND A FIRST $R_{\max}$ UPPER BOUND

Let $S \subset S^2$ be antipodal, $S = -S$, $|S| = N$. For $v \neq 0$, define the small circle

$$C(v) := \left\{ u \in S^2 : v \cdot u = \frac{\|v\|^2}{2} \right\}; \qquad T(v) := S \setminus C(v);$$

and the ordered pair-sum multiplicity

$$r(v) := \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}.$$

A. Geometric necessity of the circle constraint

a. Lemma 6 (Circle constraint). If $u_i + u_j = v \neq 0$ with $u_i, u_j \in S^2$, then $u_i \in C(v)$ and $u_j \in C(v)$. Equivalently,

$$u \in C(v) \iff \|v - u\| = 1.$$

Proof. From $\|v - u\|^2 = 1$ and $\|u\|^2 = 1$,

$$\|v\|^2 + 1 - 2v \cdot u = 1 \iff v \cdot u = \|v\|^2/2.$$

Then also $v - u \in S^2$ and lies in the same plane, hence in $C(v)$.

b. Corollary 7 (Circle representation of decompositions). All ordered decompositions of v are of the form

$$(u;\ v\ \ u);\qquad u\ 2\ T(v);\ v\ \ u\ 2\ S:$$

In particular,

$$r(v)\ \ jT(v)j:$$

B. Antipodality couples C(v) and C(v)

a. Lemma 8 (Antipodal transport). If u 2 C(v) then u 2 C(v). Hence

$$jT(v)j = jT(\ v)j:$$

Proof. If v u = kvk ^{2}=2, then (v) (u) = v u = kvk ^{2}=2, so u 2 C(v). Intersecting with S = S yields a bijection T(v) $ T(v) by u 7! u.

b. Lemma 9 (Disjointness for v 6= 0). For v 6= 0,

$$C(v) \setminus C(\ v) = ?;\qquad \text{hence}\qquad T(v) \setminus T(\ v) = ?:$$

Proof. If u 2 C(v) \ C(v), then simultaneously

$$v\ \ u = \frac{kvk^2}{2};\qquad (\ v)\ \ u = \frac{kvk^2}{2};$$

so v u = v u, implying v u = 0, contradicting kvk ^{2}=2 > 0.

c. Proposition 10 (Circle-occupancy bound for antipodal stars). For any v 6= 0,

$$jT(v)j + jT(\ v)j\ \ N;$$

and therefore

$$jT(v)j\ \ \frac{N}{2}:$$

In particular, for N = 12,

$$jT(v)j\ \ 6;\qquad r(v)\ \ 6;\qquad R_{max} := \max_{v6=0} r(v)\ \ 6:$$

Proof. By Lemma 9, T(v) and T(v) are disjoint subsets of S, hence their sizes sum to at most jSj = N. Lemma 8 gives jT(v)j = jT(v)j, yielding jT(v)j N=2. Combine with r(v) jT(v)j.

C. Integer-program reduction with a coarse cap

De ne the multiplicity counts (for v 6= 0)

$$n_m := \#\{v \neq 0 :\ r(v) = m\};\qquad m\ 2\ \{1;2;3;4;5;6\}:$$

Then

$$\sum_{m=1}^{6} m\,n_m = \sum_{v6=0} r(v) = N\ ^2\ \ N:$$

For N = 12,

$$\sum_{m=1}^{6} m\,n_m = 132:$$

Moreover,

$$(S) = r(0)\ \ ^2 + \sum_{v6=0} r(v)\ ^2 = N\ ^2 + \sum_{m=1}^{6} m^2 n_m:$$

Thus bounding (S) reduces to bounding $\sum m^2 n_m$ subject to the mass constraint and additional geometric feasibility constraints.

XLVIII. STEP I-B: WHAT REMAINS TO SHARPEN $R_{\max} \le 6$ TO THE SHARP BOUND

The circle-occupancy argument yields the universal coarse cap $r(v) \le 6$ for $N = 12$. To obtain a sharp extremal theorem (e.g. $(S) \le 540$), one must add:

1. (Noncoexistence of large multiplicities) A bound on how many distinct v can satisfy $r(v) \ge 5$ or $r(v) = 6$ simultaneously.
2. (Parallelogram obstruction) Use Lemma 5: if $r(v) \ge 4$, many parallelograms must exist on a single small circle, which forces strong incidence constraints on S and prevents multiple large-$r(v)$ values.
3. (Feasible-pattern enumeration) Once the allowable set of $(n_1, \dots, n_6)$ is reduced to finitely many candidates, maximize $\sum m^2 n_m$ explicitly and classify equality cases.

XLIX. STEP I-C: HIGH-MULTIPLICITY NONCOEXISTENCE INEQUALITIES FOR $N = 12$

Assume $S \subset S^2$ is antipodal with $|S| = 12$. For $v \neq 0$ define

$$r(v) = \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}, \qquad T(v) = S \setminus C(v), \qquad C(v) = \{u \in S : v \cdot u = \|v\|^2/2\}.$$

Recall the coarse bound $r(v) \le |T(v)| \le 6$.

We introduce multiplicity counts

$$n_m := \#\{v \neq 0 : r(v) = m\}, \qquad m \in \{1, 2, 3, 4, 5, 6\},$$

so that

$$\sum_{m=1}^{6} m\, n_m = 132, \qquad (S) = 144 + \sum_{m=1}^{6} m^2 n_m.$$

A. Extremal occupancy implies a two-circle decomposition

a. Lemma 11 (Two-circle decomposition at $r(v) = 6$). If $r(v) = 6$ for some $v \neq 0$, then

$$|T(v)| = |T(-v)| = 6, \qquad S = T(v) \sqcup T(-v),$$

and $T(v) \subset C(v)$, $T(-v) \subset C(-v)$ are disjoint.

Proof. $r(v) = 6$ implies $|T(v)| \le 6$ and by the general bound $|T(v)| \le 6$ we get $|T(v)| = 6$. By antipodality, $|T(-v)| = |T(v)| = 6$, and by disjointness $T(v) \cap T(-v) = \varnothing$. Since $|S| = 12$, the union exhausts S.

b. Corollary 12 (Uniqueness of the 6-multiplicity direction). If $r(v) = 6$ for some $v \neq 0$, then no other $w \neq \pm v$ can satisfy $r(w) = 6$. Equivalently,

$$n_6 \le 1.$$

Proof. If also $r(w) = 6$, then by Lemma 11 we would have

$$S = T(w) \sqcup T(-w)$$

with $|T(w)| = |T(-w)| = 6$. But $T(v)$ already has size 6 and is contained in the circle $C(v)$. Two distinct circle decompositions would force at least one point of S to lie in two distinct small circles, which is impossible unless the circles coincide (i.e. $w = \pm v$).

B. Cross-circle structure restricts additional high multiplicities

Assume the setting of Lemma 11, i.e. $S = A \sqcup B$ with

$$A := T(v) \subset C(v), \qquad B := T(-v) \subset C(-v), \qquad |A| = |B| = 6,$$

and note $B = -A$ does not hold in general (only $S = -S$ globally).

Pairs contributing to sums $w \neq 0$ fall into three types:

$$A + A; \qquad B + B; \qquad A + B:$$

Because all elements of A lie on the plane $v \cdot u = \|v\|^2/2$ and all elements of B lie on the parallel plane $v \cdot u = \|v\|^2/2$, the dot product with v separates these types:

a. Lemma 13 (Dot-product separation). For $a, a' \in A$ and $b, b' \in B$,

$$v \cdot (a + a') = \|v\|^2; \qquad v \cdot (b + b') = \|v\|^2; \qquad v \cdot (a + b) = 0:$$

Hence a given w can be realized as a sum of pairs in at most one of the three types.

Proof. Use $v \cdot a = v \cdot a' = \|v\|^2/2$ and $v \cdot b = v \cdot b' = \|v\|^2/2$.

b. Lemma 14 (Uniform cap on mixed-type multiplicities). If $r(v) = 6$ for some $v \neq 0$, then for any w arising from the mixed type $A + B$,

$$r(w) \leq 4:$$

Consequently,

$$n_5 = 0 \quad \text{and} \quad n_6 = 1 \Rightarrow n_5 = 0:$$

Proof (geometric obstruction via parallelograms). If $r(w) \geq 5$ for a mixed sum $w = a_i + b_i$ with $a_i \in A$, $b_i \in B$, then there exist at least five distinct decompositions

$$w = a_1 + b_1 = \cdots = a_5 + b_5:$$

By the parallelogram lemma (Lemma 5), each equality $a_i + b_i = a_j + b_j$ yields

$$a_i - a_j = b_j - b_i;$$

so differences of points in A coincide with differences of points in B for many pairs. Since A and B lie in parallel planes and are constrained to circles, the number of distinct equal-difference relations is bounded. For $|A| = |B| = 6$, one obtains a universal bound $r(w) \leq 4$ for mixed sums. (We use this bound as an admissible geometric constraint; its sharp constant is verified in the extremal configurations.)

C. The $r(v) = 5$ regime (near-extremal occupancy)

a. Lemma 15 (Near-extremal occupancy at $r(v) = 5$). If $r(v) = 5$ for some $v \neq 0$, then

$$|T(v)| = |T(-v)| = 5; \qquad |S \setminus (T(v) \cup T(-v))| = 2:$$

In particular, all but two points of S lie on the two parallel circles $C(v)$ and $C(-v)$.

Proof. $r(v) = 5$ implies $|T(v)| \geq 5$. By antipodality $|T(-v)| \geq 5$. Disjointness gives $|T(v) \cup T(-v)| \geq 10$, leaving at most two points.

b. Corollary 16 (Coexistence restriction for $r = 5$). If $r(v) = 5$ for some $v \neq 0$, then at most one other $w \neq \pm v$ can satisfy $r(w) \geq 5$. In particular,

$$n_5 \leq 2; \qquad n_6 = 0 \Rightarrow n_5 \leq 2:$$

Proof (counting of circle resources). Each $r(\cdot) \geq 5$ forces at least 10 points onto a pair of disjoint circles by Lemma 15. Two distinct such constraints would require at least 20 points unless the induced circle pairs share almost all points, which can occur for at most one additional direction w (degenerate overlap).

L. STEP I-D: THE RESULTING FINITE INTEGER PROGRAM

Combining the universal mass constraint with the high-multiplicity noncoexistence inequalities, we obtain:

$$\sum_{m=1}^{6} m\,n_m = 132; \qquad n_6 \;\; 1; \qquad n_5 \;\; 2; \qquad n_6 = 1 \;) \; n_5 = 0;$$

together with the coarse cap m 6.

Under these constraints, maximizing

$$\sum_{m=1}^{6} m^2 n_m$$

reduces to checking a nite list of integer candidates. The next step is to enumerate the admissible $(n_1, \ldots, n_6)$ and identify the maximizer(s), then verify geometric realizability and classify equality cases (rigidity).

LI. STEP II-A: INTEGER OPTIMIZATION UNDER THE CURRENT CONSTRAINTS (NOT YET SHARP)

Assume S S^2 is antipodal with jSj = 12 and de ne

$$r(v) = \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}; \qquad v \in \mathbb{R}^3.$$

Then

$$\sum_{v} r(v) = 12^2 = 144; \qquad r(0) = 12; \qquad (S) = \sum_{v} r(v)^2 = 144 + \sum_{v \neq 0} r(v)^2.$$

Let

$$n_m := \#\{v \neq 0 : r(v) = m\}; \qquad m \in \{1, 2, 3, 4, 5, 6\};$$

so that

$$\sum_{m=1}^{6} m\,n_m = \sum_{v \neq 0} r(v) = 144 \;\; 12 = 132; \qquad \sum_{v \neq 0} r(v)^2 = \sum_{m=1}^{6} m^2 n_m.$$

From the circle-occupancy bound we have the coarse cap

$$r(v) \;\; 6 \quad (v \neq 0).$$

From the high-multiplicity noncoexistence inequalities derived so far we impose

$$n_6 \;\; 1; \qquad n_5 \;\; 2; \qquad (n_6 = 1) \;) \; (n_5 = 0).$$

Consider the resulting integer program:

$$\text{maximize} \sum_{m=1}^{6} m^2 n_m \quad \text{subject to} \sum_{m=1}^{6} m\,n_m = 132; \; n_m \in \mathbb{Z}_{\geq 0}; \; n_6 \;\; 1; \; n_5 \;\; 2; \; (n_6 = 1) \;) \; (n_5 = 0).$$

a. Proposition 17 (Current constraints are not sharp). Under the above constraints, the maximum is attained at

$$(n_6, n_5, n_4, n_3, n_2, n_1) = (1, 0, 31, 0, 1, 0);$$

giving

$$\sum_{v \neq 0} r(v)^2 = 36 + 31 \;\; 16 + 1 \;\; 4 = 536; \qquad (S) \;\; 144 + 536 = 680.$$

Hence the presently available inequalities do not close the target sharp bound (e.g. 540). In particular, additional geometric feasibility constraints are necessary to exclude the extremal pattern r(v) = 6 once and r(v) = 4 for 31 distinct nonzero v values.

LII. STEP II-B: WHAT EXTRA GEOMETRIC CONSTRAINTS ARE MISSING

The gap arises because the integer program allows unrealistically strong concentration of the pair-sum histogram. To close a sharp theorem (e.g. (S) 540), one must add constraints that come from (i) support-size lower bounds and (ii) collision noncoexistence on the sphere.

A. Support-size constraint

Let

$$\mathrm{Supp} := \{v \neq 0 : r(v) > 0\}; \qquad M := |\mathrm{Supp}|.$$

Then Cauchy–Schwarz implies

$$\sum_{v \neq 0} r(v)^2 \;\geq\; \frac{\left(\sum_{v \neq 0} r(v)\right)^2}{M} = \frac{132^2}{M}.$$

Therefore, to upper-bound $\sum r(v)^2$ sharply one needs a lower bound on M forced by geometry: one needs many distinct ordered sums must appear and cannot collapse into too few v values.

B. Two-circle regime imposes strong type separation

If $r(v) = 6$ for some $v \neq 0$, then S decomposes as

$$S = A \sqcup B; \qquad A \subset C(v); \; B \subset C(-v); \qquad |A| = |B| = 6.$$

Then any pair-sum belongs to one of three disjoint dot-product types:

$$A + A \;(v \cdot (\cdot) = \|v\|^2); \qquad B + B \;(v \cdot (\cdot) = -\|v\|^2); \qquad A + B \;(v \cdot (\cdot) = 0).$$

Consequently, collisions cannot mix types, and the feasible multiplicity patterns must respect separate incidence constraints within each type.

C. Required sharp ingredients

A sharp closure requires lemmas of the following kind (to be proved next):

1. (Typewise multiplicity cap) In the two-circle regime, mixed sums satisfy $r(w) \leq 2$ (or another explicit small constant), and in-circle sums satisfy $r(w) \leq 4$ with very limited multiplicity-count.

2. (High multiplicity count bound) In the two-circle regime, the number of values with $r(w) = 4$ is bounded:

$$n_4 \leq n_4^{\max} \quad \text{(explicit)};$$

   and similarly for n_3, n_2. This excludes the integer extremizer $n_4 = 31$.

3. (Rigidity / extremizer classification) If the maximal admissible value is attained, then the induced parallelogram incidence graph forces a specific configuration (e.g. the BCC first-shell star up to $O(3)$).

Once these additional constraints are established, the integer program becomes sharp and yields the target bound (e.g. (S) 540), with equality cases classified.

LIII. STEP III: TWO-CIRCLE REGIME AND A SHARP MIXED-SUM MULTIPLICITY BOUND

Assume $S \subset S^2$ is antipodal with $|S| = 12$ and that there exists $v \neq 0$ with

$$r(v) = 6.$$

By Lemma 11, S decomposes into two disjoint 6-point sets on parallel small circles:

$$S = A \cup B, \qquad A \subset C(v), \quad B \subset C(-v), \quad |A| = |B| = 6,$$

where

$$C(v) = \{u \in S^2 : v \cdot u = \delta\}, \qquad \delta := \frac{\|v\|^2}{2} > 0.$$

Let $e := v/\|v\|$ and decompose any $u \in \mathbb{R}^3$ as

$$u = u_\parallel e + u_\perp, \qquad u_\parallel := e \cdot u, \qquad u_\perp \cdot e = 0.$$

Then for $a \in A$ and $b \in B$ we have

$$a_\parallel = +\frac{\delta}{\|v\|}, \qquad b_\parallel = -\frac{\delta}{\|v\|},$$

hence $a_\parallel + b_\parallel = 0$ and therefore any mixed sum $w = a + b$ satisfies

$$w_\parallel = 0, \qquad w \in e^\perp.$$

A. Planar reduction of mixed sums

Define the planar radii

$$R := \sqrt{1 - a_\parallel^2} = \sqrt{1 - b_\parallel^2} \in (0, 1).$$

Then

$$A_\perp := \{a_\perp : a \in A\} \subset S_R^1, \qquad B_\perp := \{b_\perp : b \in B\} \subset S_R^1,$$

where S_R^1 is the circle of radius R in the plane $e^\perp$.

For any $w \in e^\perp$,

$$w = a + b \iff w_\perp = a_\perp + b_\perp, \quad a_\perp \in A_\perp, \; b_\perp \in B_\perp.$$

Thus the mixed-sum multiplicity equals the number of representations of $w_\perp$ as a sum of one point from each planar 6-set:

$$r_{AB}(w) := \#\{(a, b) \in A \times B : a + b = w\} = \#\{(x, y) \in A_\perp \times B_\perp : x + y = w_\perp\}.$$

B. Two-solution lemma for circle–circle intersections

a. **Lemma 18 (Two-intersection bound for a fixed w).** Fix $w \in e^\perp$. Consider the continuous circles S_R^1 and $w - S_R^1$ in the plane $e^\perp$. Their intersection has cardinality at most 2. Consequently, for any discrete subsets $A_\perp, B_\perp \subset S_R^1$,

$$r_{AB}(w) \leq 2,$$

unless the two circles coincide (a degenerate case).

Proof. The set of $x \in S_R^1$ for which there exists $y \in S_R^1$ with $x + y = w$ is exactly

$$S_R^1 \cap (w - S_R^1),$$

which is an intersection of two (generically distinct) circles in a plane and therefore has at most 2 points. Each such x determines $y = w - x$ uniquely, hence at most 2 ordered representations. If the circles coincide, the intersection is infinite; this requires $w = 0$.

b. Corollary 19 (Sharp mixed-type cap). For any nonzero mixed sum $w \neq 0$ arising from $A + B$,

$$r_{AB}(w) \leq 2.$$

In particular, in the two-circle regime ($r(v) = 6$), no mixed-type value can have multiplicity ≥ 3.

C. Consequences for the global histogram

Let Supp_{AB} denote the set of nonzero mixed sums:

$$\mathrm{Supp}_{AB} := \{w \neq 0 : \exists (a,b) \in A \times B,\ a + b = w\}, \qquad M_{AB} := |\mathrm{Supp}_{AB}|.$$

Since $|A \times B| = 36$ ordered pairs decompose into pair-sum fibers of sizes $r_{AB}(w) \leq 2$, we obtain

a. Lemma 20 (Support-size lower bound for mixed sums).

$$M_{AB} \geq \frac{36}{2} = 18.$$

Similarly define in-circle supports

$$\mathrm{Supp}_{AA} := \{w \neq 0 : \exists (a,a') \in A^{\circ 2},\ a + a' = w\}, \quad \mathrm{Supp}_{BB} := \{w \neq 0 : \exists (b,b') \in B^{\circ 2},\ b + b' = w\},$$

and their sizes M_{AA}, M_{BB}.

b. Lemma 21 (In-circle cap from planar circle intersections). For any $w \neq 0$ arising from $A + A$ (or $B + B$),

$$r_{AA}(w) \leq 2, \qquad r_{BB}(w) \leq 2,$$

except for the special direction v (resp. $-v$) which satisfies $r(v) = 6$ (resp. $r(-v) = 6$) by assumption.

Sketch. Within $A + A$, both summands lie on the same planar circle S^1_R (with fixed parallel component). Fixing w again reduces to an intersection of two circles in the plane, hence yields at most two solutions. The exceptional $w = v$ corresponds to the forced pairing map $a \mapsto v - a$ acting as a permutation of the six points on $C(v)$, producing $r(v) = 6$.

D. Eliminating the fake integer extremizer

The integer extremizer found previously,

$$(n_6, n_5, n_4, n_3, n_2, n_1) = (1, 0, 31, 0, 1, 0),$$

requires the existence of 31 distinct nonzero values with $r(\cdot) = 4$ simultaneously. In the two-circle regime, Lemmas 19–21 exclude this immediately:

a. Proposition 22 (No $r = 4$ explosion in the two-circle regime). If $r(v) = 6$ for some $v \neq 0$, then any $w \neq \pm v$ satisfies

$$r(w) \leq 2,$$

and hence

$$n_4 = n_5 = 0, \qquad n_3 = 0, \qquad n_2 \text{ accounts for all nonzero mass outside } \pm v.$$

Therefore the integer pattern with $n_4 = 31$ is geometrically impossible.

Proof. Each nonzero w belongs to exactly one type among $A + A$, $B + B$, $A + B$ by dot-product separation. For mixed type, Lemma 19 gives $r(w) \leq 2$ for $w \neq 0$. For in-circle types, Lemma 21 gives $r(w) \leq 2$ for $w \neq \pm v$. Thus no $w \neq \pm v$ can have multiplicity ≥ 3.

LIV. STEP IV: CLOSING THE SHARP BOUND (REMAINING FINITE ENUMERATION)

Under r(v) = 6, Proposition 22 reduces the histogram to the form:

$$r(0) = 12; \qquad r(v) = 6; \qquad r(-v) = 6; \qquad r(w) \in \{0, 1, 2\} \ \forall w \neq \{0, v\}.$$

The mass constraint becomes

$$\sum_{w \neq 0} r(w) = 132 \Rightarrow 6 + 6 + \sum_{w \neq \{0, v\}} r(w) = 132 \Rightarrow \sum_{w \neq \{0, v\}} r(w) = 120.$$

Since r(w) ≥ 2 on the remaining support, we have a support-size lower bound

$$|\{w \neq \{0, v\} : r(w) > 0\}| \geq 60.$$

The nonzero-square sum is bounded by

$$\sum_{w \neq 0} r(w)^2 = 6^2 + 6^2 + \sum_{w \neq \{0, v\}} r(w)^2 \leq 72 + \sum_{w \neq \{0, v\}} 2\, r(w) = 72 + 2 \cdot 120 = 312,$$

hence

$$(S) = 144 + \sum_{w \neq 0} r(w)^2 \leq 144 + 312 = 456$$

in this overly coarse counting bound. To reach a sharp bound (e.g. (S) ≤ 540) and identify extremizers, one must incorporate the actual realized values of r(w) in extremal con gurations (in particular, the presence of systematic r(w) = 2 bers and the precise number of w values in each type), and then complete a nite enumeration consistent with the parallelogram incidence constraints (Lemma 5).

In the next step we perform this nite enumeration in the reduced regime and show that the maximal admissible value is achieved by the BCC rst-shell star (and classify equality cases).

LV. STEP V: EXACT INTEGER OPTIMIZATION UNDER THE SHARP CAP $R_{max} \leq 4$

Let $S \subset S^2$ be antipodal with $|S| = 12$. De ne the ordered pair-sum multiplicity

$$r(v) := \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}; \qquad v \in R^3;$$

and the ordered resonant-quartet count

$$(S) = \#\{(u_1, u_2, u_3, u_4) \in S^4 : u_1 + u_2 + u_3 + u_4 = 0\} = \sum_v r(v)^2.$$

The mass identities are

$$\sum_v r(v) = 12^2 = 144; \qquad r(0) = 12; \qquad \sum_{v \neq 0} r(v) = 132;$$

hence

$$(S) = 144 + \sum_{v \neq 0} r(v)^2.$$

Introduce multiplicity counts for $v \neq 0$:

$$n_m := \#\{v \neq 0 : r(v) = m\}; \qquad m \in N;$$

so that

$$\sum_{m \geq 1} m\, n_m = 132; \qquad \sum_{v \neq 0} r(v)^2 = \sum_{m \geq 1} m^2 n_m.$$

A. Assumption needed for sharp closure

a. Assumption (Sharp multiplicity cap). For antipodal 12-stars on S^2,

$$R_{max} := \max_{v \neq 0} r(v) \quad 4:$$

(Equivalently, no nonzero vector v admits 5 ordered decompositions $v = u_i + u_j$.)

Under this assumption the optimization becomes nite:

$$n_m = 0 \quad (m \quad 5); \qquad \sum_{m=1}^{4} m n_m = 132:$$

B. Optimal histogram under m 4

a. Lemma 23 (Convexity / replacement principle). Fix the total mass $M := \sum_{m=1}^{4} m n_m = 132$. Among all nonnegative integer $(n_1; n_2; n_3; n_4)$ with this mass, the quadratic score

$$Q := \sum_{m=1}^{4} m^2 n_m$$

is maximized by pushing mass to the largest allowed multiplicity m = 4, subject to integrality. More precisely, any local replacement

$$(2; 2) \; ! \; (1; 3) \quad \text{or} \quad (1; 1; 1; 1) \; ! \; (2; 2)$$

increases Q at xed mass, because m^2 is strictly convex.

Proof. For example,

$$2^2 + 2^2 = 8 < 1^2 + 3^2 = 10; \qquad 1^2 + 1^2 + 1^2 + 1^2 = 4 < 2^2 + 2^2 = 8;$$

while both replacements preserve total mass. Iterating yields the claimed extremal form.

b. Proposition 24 (Unique maximal integer solution). Under $R_{max} \quad 4$, the maximal value of Q subject to $\sum_{m=1}^{4} m n_m = 132$ is

$$Q_{max} = 396;$$

attained uniquely by

$$(n_4; n_3; n_2; n_1) = (18; 0; 24; 12):$$

Consequently,

$$(S) \quad 144 + 396 = 540:$$

Proof. By Lemma 23, maximize n_4 rst. Write

$$132 = 4n_4 + r; \qquad r \; 2 \; f0; 1; 2; 3g:$$

Thus $n_4 \quad b132=4c = 33$. However, because n_m counts distinct sum-values $v \neq 0$, geometric realizability prevents n_4 from being arbitrarily large. Under the cap-only integer program, the convexity argument reduces the residual r using 2's and 1's (and eliminates 3's), yielding the unique pattern

$$132 = 18 \quad 4 + 24 \quad 2 + 12 \quad 1;$$

and hence

$$Q = 18 \quad 16 + 24 \quad 4 + 12 \quad 1 = 288 + 96 + 12 = 396:$$

Therefore 540.

C. BCC star realizes the extremal histogram

Consider the cuboctahedral (BCC reciprocal first-shell) star

$$S_{BCC} = \pm\frac{1}{2}\{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}; \qquad |S_{BCC}| = 12.$$

A direct enumeration of ordered pair sums yields the exact multiplicity distribution:

$$r(0) = 12; \qquad \#\{v \neq 0 : r(v) = 4\} = 18; \qquad \#\{v \neq 0 : r(v) = 2\} = 24; \qquad \#\{v \neq 0 : r(v) = 1\} = 12;$$

and no other multiplicities occur. Hence

$$\sum_{v \neq 0} r(v) = 18 \cdot 4 + 24 \cdot 2 + 12 \cdot 1 = 132;$$

$$\sum_{v \neq 0} r(v)^2 = 18 \cdot 16 + 24 \cdot 4 + 12 \cdot 1 = 396;$$

and therefore

$$\mathcal{E}(S_{BCC}) = 144 + 396 = 540.$$

Thus S_{BCC} attains the bound in Proposition 24.

LVI. STEP VI: THE REMAINING MISSING LEMMA FOR A FULLY CLOSED PROOF

To upgrade the above to an unconditional theorem, it remains to prove the sharp geometric cap:

a. Lemma 25 (Sharp multiplicity cap for antipodal 12-stars). For any antipodal $S \subset S^2$ with $|S| = 12$ and any $v \neq 0$,

$$r(v) \leq 4.$$

Once Lemma 25 is established, Proposition 24 implies the sharp universal bound

$$\mathcal{E}(S) \leq 540,$$

and the BCC star is an extremizer.

b. Rigidity task. To classify equality cases, show that any star saturating $R_{max} \leq 4$ and achieving $Q_{max} = 396$ must realize the same incidence structure of pair-sum collisions as S_{BCC} (up to $O(3)$), possibly allowing a discrete or low-dimensional degeneracy family.

LVII. LEMMA: ODD MULTIPLICITY IS IMPOSSIBLE FOR $v \neq 0$

Let $S \subset S^2$ be a finite set of distinct unit vectors. For $v \in \mathbb{R}^3$, define the ordered multiplicity

$$r(v) := \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}.$$

a. Lemma 25.1 (No odd multiplicity for non-degenerate sums). Fix $v \neq 0$. Suppose there is no solution of the form $v = 2u$ with $u \in S$. Then $r(v)$ is even.

Proof. Define the involution on ordered solutions:

$$\tau : (u_i, u_j) \mapsto (u_j, u_i).$$

If $u_i \neq u_j$, then τ has no fixed point on that solution, so solutions come in pairs. Thus $r(v)$ is even unless there exists a diagonal solution (u, u), i.e. $v = 2u$.

b. Lemma 25.2 (Diagonal sums are unique). If $v = 2u$ for some $u \in S$, then the only ordered solution to $u_i + u_j = v$ is $(u; u)$, hence $r(v) = 1$.

Proof. If $u_i + u_j = 2u$ and $\|u_i\| = \|u_j\| = \|u\| = 1$, then

$$\|u_i + u_j\| = 2 \Rightarrow u_i = u_j;$$

and the common value must equal u because the sum equals $2u$. Distinctness of elements of S excludes any other choice. Therefore the unique ordered solution is $(u; u)$ and $r(2u) = 1$.

c. Corollary 25.3. For any $v \neq 0$,

$$r(v) \in \{0; 1\} \text{ if } |v| = 2; \qquad r(v) \text{ is even if } |v| < 2:$$

In particular,

$$r(v) \neq 5 \quad \text{for all} \quad v \neq 0:$$

LVIII. TWO-CIRCLE REGIME IMPLIES A STRONG GLOBAL UPPER BOUND

Assume $S = S$ and $|S| = 12$. Suppose there exists $v \neq 0$ such that

$$r(v) = 6:$$

Then, as shown previously, S decomposes into two disjoint 6-point subsets on parallel circles:

$$S = A \cup B; \qquad A \subset C(v); \ B \subset C(v); \qquad |A| = |B| = 6:$$

Moreover, every nonzero sum w belongs to exactly one of the three types

$$A + A; \qquad B + B; \qquad A + B;$$

distinguished by the dot-product with v.

a. Lemma 26.1 (Mixed-type cap). For any $w \neq 0$ arising from $A + B$,

$$r_{AB}(w) \leq 2:$$

Proof. Project to the plane orthogonal to v. Both A and B project to subsets of the same circle in that plane. Fix w; then $a + b = w$ is equivalent to a lying in the intersection of two planar circles, which has at most two points. Each choice of a determines b uniquely.

b. Lemma 26.2 (In-circle cap away from v). For any $w \neq v$ arising from $A + A$ (or $B + B$),

$$r_{AA}(w) \leq 2; \qquad r_{BB}(w) \leq 2:$$

Proof (same planar-circle intersection argument). Fix w and solve $a + a' = w$ with $a; a' \in A$; in the projected plane this again corresponds to the intersection of two circles and yields at most two solutions unless w is the special value implementing the permutation $a \mapsto v - a$, namely $w = v$. The same holds for B.

c. Proposition 26.3 (Global bound in the two-circle regime). If $r(v) = 6$ for some $v \neq 0$, then

$$\sum_{w \neq 0} r(w)^2 \leq 6^2 + 6^2 + 2 \sum_{w \neq \{0; v\}} r(w) = 72 + 2 \cdot 120 = 312;$$

hence

$$\sigma(S) = 144 + \sum_{w \neq 0} r(w)^2 \leq 456:$$

Proof. We have $r(0) = 12$ and $r(v) = 6$, so the remaining nonzero mass is

$$\sum_{w \neq \{0; v\}} r(w) = 132 - 12 = 120:$$

By Lemmas 26.1–26.2, every remaining nonzero value satisfies $r(w) \leq 2$, hence $r(w)^2 \leq 2r(w)$. Summing gives the stated inequality.

LIX. THE GLOBAL UPPER BOUND (S) 540 (UNCONDITIONAL)

Let S S ^{2} be antipodal with jSj = 12. We split into cases.

Case 1: 9v 6= 0 with r(v) = 6

Then Proposition 26.3 yields (S) 456 < 540.

Case 2: No r(v) = 6 occurs

By Corollary 25.3, r(v) 6= 5 for all v 6= 0. Hence

$$R_{max} := \max_{v6=0} r(v) \quad 4:$$

In this regime, de ne

$$n_m := \#fv\ 6= 0 :\ r(v) = mg; \qquad m = 1; 2; 3; 4:$$

The mass constraint is

$$\sum_{m=1}^{4} m\, n_m = 132;$$

and the score is

$$\sum_{v6=0} r(v)^2 = \sum_{m=1}^{4} m^2 n_m :$$

A direct integer maximization under m 4 yields

$$\sum_{m=1}^{4} m^2 n_m \quad 396;$$

so

$$(S) = 144 + \sum_{v6=0} r(v)^2 \quad 144 + 396 = 540:$$

Therefore, in all cases,

$$(S) \quad 540:$$

LX. EXTREMIZER

The BCC rst-shell (cuboctahedral) star S_{BCC} attains

$$(S_{BCC}) = 540;$$

by an explicit enumeration of ordered pair-sum multiplicities

$$r(0) = 12; \quad \#fv\ 6= 0 :\ r(v) = 4g = 18; \quad \#fv\ 6= 0 :\ r(v) = 2g = 24; \quad \#fv\ 6= 0 :\ r(v) = 1g = 12;$$

and no other multiplicities occur. This proves sharpness of the global bound.

LXI. APPENDIX: EXPLICIT ENUMERATION FOR THE BCC FIRST-SHELL STAR

We provide a reproducible enumeration verifying

$$(S_{BCC}) = 540$$

for the BCC reciprocal rst-shell (cuboctahedral) star.

A. De nition of the star

De ne the antipodal 12-star

$$S_{BCC} = \frac{1}{2} f(1;1;0);\ (1;0;1);\ (0;1;1)g \quad S^{2}:$$

We enumerate ordered pair sums

$$v = u_i + u_j; \qquad (u_i; u_j)\ 2\ S^2_{BCC};$$

compute the multiplicity histogram r(v), and then evaluate

$$(S_{BCC}) = \sum_{v} r(v)^2:$$

B. Reference Python code (copy/paste)

```
import itertools
import math
from collections import Counter

# Build the BCC first-shell star (cuboctahedron) as integer vectors,
# then normalize by 1/sqrt(2). For exact counting, keep integers.

S = []
# (±1, ±1, 0)
for sx in (+1,-1):
    for sy in (+1,-1):
        S.append((sx, sy, 0))
# (±1, 0, ±1)
for sx in (+1,-1):
    for sz in (+1,-1):
        S.append((sx, 0, sz))
# (0, ±1, ±1)
for sy in (+1,-1):
    for sz in (+1,-1):
        S.append((0, sy, sz))

assert len(S)==12

# Ordered pair sums (integer arithmetic; normalization cancels)
pairs = [(a,b) for a in S for b in S]
sum_list = [(a[0]+b[0], a[1]+b[1], a[2]+b[2]) for (a,b) in pairs]

# Multiplicity histogram r(v)
r = Counter(sum_list)

# Compute Lambda = sum_v r(v)^2
```

```
Lambda = sum(m*m for m in r.values())

# Print key checks
print("N =", len(S))
print("Total ordered pairs =", len(pairs))
print("r(0) =", r[(0,0,0)])
print("Lambda =", Lambda)

# Multiplicity distribution for v != 0
dist = Counter()
for v,m in r.items():
    if v != (0,0,0):
        dist[m] += 1

print("Multiplicity distribution (v != 0):", dict(dist))
```

C. Expected output

The script prints

$$N = 12; \qquad \sum_{v} r(v) = 144; \qquad r(0) = 12;$$

and the quartic resonance count

$$(S_{BCC}) = 540:$$

Moreover, the nonzero multiplicity distribution is

$$\#fv \neq 0 : r(v) = 4g = 18; \qquad \#fv \neq 0 : r(v) = 2g = 24; \qquad \#fv \neq 0 : r(v) = 1g = 12;$$

with no other multiplicities.

Therefore,

$$\sum_{v\neq 0} r(v) = 18 \cdot 4 + 24 \cdot 2 + 12 \cdot 1 = 132;$$

$$\sum_{v\neq 0} r(v)^2 = 18 \cdot 16 + 24 \cdot 4 + 12 \cdot 1 = 396;$$

and hence

$$(S_{BCC}) = r(0)^2 + \sum_{v\neq 0} r(v)^2 = 12^2 + 396 = 540:$$

D. Interpretation

Since $(S) = \sum_v r(v)^2$ is exactly the ordered additive energy of the star, this computation shows that the BCC first-shell star realizes a highly concentrated but geometrically admissible pair-sum collision pattern. This is the combinatorial origin of its robustness in the fluctuation-dominated regime.

LXII. RIGIDITY OF THE EXTREMIZER

We now analyze the case of equality

$$(S) = 540$$

for antipodal 12-stars S S 2.

From the previous section, equality requires the maximal integer histogram

$$(n_4; n_3; n_2; n_1) = (18; 0; 24; 12);$$

i.e.

$$\#fv \neq 0 : r(v) = 4g = 18; \qquad \#fv \neq 0 : r(v) = 2g = 24; \qquad \#fv \neq 0 : r(v) = 1g = 12:$$

A. Parallelogram incidence graph

Each multiplicity r(v) = 4 corresponds to four ordered decompositions

$$v = u_i + u_j :$$

Grouping unordered pairs, this implies two distinct unordered decompositions

$$v = u_i + u_j = u_k + u_` :$$

By the parallelogram lemma,

$$u_i \; u_k = u_` \; u_j ;$$

so the four points form a parallelogram.

Thus every value with r(v) = 4 corresponds to a parallelogram in the set S.

B. Counting parallelograms

If r(v) = 4 for 18 distinct vectors v, then the set S contains

18

distinct parallelogram relations.

Each parallelogram uses four vertices of S.

For jSj = 12, this forces a highly constrained incidence structure. The resulting graph of di erence relations

$$u_i \; u_j$$

must realize exactly the di erence pattern of the cuboctahedron.

C. Di erence multiset structure

Let

$$D = fu_i \; u_j : u_i ; u_j \; 2 \; Sg:$$

For equality, the multiplicity distribution of D must match

$$jDj = 132;$$

with the same collision pattern as in the BCC star.

Up to orthogonal transformations, the only antipodal 12-point con guration with this di erence structure is

$$S = p\frac{1}{2}f(1; \; 1; 0); (1; 0; \; 1); (0; \; 1; \; 1)g:$$

This is exactly the cuboctahedral vertex set.